



TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING

A
MINOR PROJECT REPORT
ON
**“AIR POLLUTION FORECASTING SYSTEM USING ENSEMBLE
LEARNING AND LSTM”**

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Lalitpur, Nepal
March 16, 2026

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Acknowledgments

In the course of completing this project, we had the opportunity to learn and apply many concepts that greatly contributed to our academic development. The successful completion of this report would not have been possible without the guidance, support, and encouragement of several individuals and institutions.

First and foremost, we would like to express our sincere gratitude to our respected supervisor, **Dr. Kishor Kumar Adhikari**, for his valuable guidance, insightful suggestions, and continuous support throughout the development of this project. His constructive feedback, patience, and encouragement helped us stay focused and motivated during the entire process. His expertise and mentorship played a significant role in shaping this work and improving its overall quality.

We would also like to extend our appreciation to the **Department of Electronics and Computer Engineering** for providing us with the academic environment and resources necessary to carry out this project. The knowledge and support provided by the faculty members of the department greatly assisted us in completing this work.

Finally, we would like to thank our friends and classmates for their encouragement, discussions, and cooperation during different stages of this project. Their support made the learning process more engaging and productive.

Abstract

Ground-based air quality monitoring in Nepal is severely limited, with fewer than 10 operational $\text{PM}_{2.5}$ sensors covering the entire country. This study investigates whether satellite remote sensing data, combined with ERA5 meteorological reanalysis, can serve as a viable alternative for near real-time $\text{PM}_{2.5}$ estimation and 7-day forecasting across geographically diverse locations in Nepal. A modular machine learning framework was developed integrating observations from three satellite sources: MODIS aerosol optical depth, Sentinel-2 multispectral bands, and Sentinel-5P tropospheric trace gases, together with ERA5 reanalysis variables. Because satellite observations are often unavailable due to cloud cover and ERA5 data has a 7–8 day publication delay, predictive gap-filling models were trained for each data source to ensure continuous daily feature availability. An LSTM-based ERA5 forecaster and an XGBoost rain classifier address the ERA5 latency, while separate gradient-boosted models fill missing satellite retrievals for MODIS, Sentinel-2, and Sentinel-5P. Transfer learning was applied for MODIS gap-filling at deployment sites, improving performance at three of four locations. The complete synthesised feature set was then used to train a stacking ensemble model for daily $\text{PM}_{2.5}$ prediction and a bidirectional LSTM model for 7-day forecasting, evaluated at four deployment sites: Kathmandu, Pokhara, Chitwan, and Birgunj. The base ensemble model achieved $R^2 = 0.8201$ and $\text{RMSE} = 11.09 \mu\text{g}/\text{m}^3$ on a chronological test set, improving to $R^2 = 0.8909$ after deployment-site fine-tuning, while the LSTM forecaster achieved $R^2 = 0.8335$ and 80.75% accuracy over the 7-day horizon, with accuracy gradually declining as lead time increased. Overall, the results demonstrate that satellite-derived features, when systematically gap-filled and combined with meteorological reanalysis, can produce reliable $\text{PM}_{2.5}$ estimates without dense ground monitoring infrastructure, offering a scalable and cost-effective approach for air quality monitoring in data-sparse regions.

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List of Abbreviations

Abbreviation	Full Form
AI (Aerosol)	Aerosol Index
AI	Artificial Intelligence
AOD	Aerosol Optical Depth
API	Application Programming Interface
AQI	Air Quality Index
CAMS	Copernicus Atmosphere Monitoring Service
CO	Carbon Monoxide
CSV	Comma Separated Value
CV	Cross Validation
DL	Deep Learning
DPT	Dew-Point Temperature
ECMWF	European Centre for Medium-Range Weather Forecasts
ERA5	ECMWF Reanalysis 5
GEE	Google Earth Engine
HCHO	Formaldehyde
HG-LSTM	Hierarchical geospatial long short-term memory
IDW	Inverse Distance Weighting
LGBM	Light Gradient Boosting Machine
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MPE	Mean Percentage Error
MERRA-2	Modern-Era Retrospective analysis for Research and Applications V2
MODIS	Moderate Resolution Imaging Spectroradiometer
NASA	National Aeronautics and Space Administration
NDVI	Normalized Difference Vegetation Index
netCDF	network Common Data Form

Abbreviation	Full Form
NO ₂	Nitrogen Dioxide
O ₃	Ozone
OpenAQ	Open Air Quality
PM _{2.5}	Particulate Matter smaller than 2.5 micrometers
R ²	Coefficient of Determination
RF	Random Forest
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
RTP	Remote Transported Pollutants
SO ₂	Sulfur Dioxide
SP	Surface Pressure
SR	Surface Reflectance
STRI	Spatio-Temporal Representation Input
STRI_fe	Spatio-Temporal Representation Input – feature extractor
STRI_p	Spatio-Temporal Representation Input – predictor
TEMP	Temperature
TP	Total Precipitation
TROPOMI	Tropospheric Monitoring Instrument
UI/UX	User Interface / User Experience
WD	Wind Direction
WS	Wind Speed
XGBoost	Extreme Gradient Boosting
YRD	Yangtze River Delta

1. Introduction

Air pollution has become one of the most significant environmental and public health challenges worldwide, particularly in developing countries experiencing rapid urbanization, industrial growth, and increasing transportation activities. In Nepal, deteriorating air quality has emerged as a major threat to public health and environmental sustainability, with fine particulate matter ($\text{PM}_{2.5}$) being one of the most harmful pollutants. Accurate monitoring and prediction of air pollution are therefore crucial for effective policy formulation, health protection, and environmental management. However, Nepal's limited and inconsistent ground-based monitoring infrastructure restricts the availability of reliable air quality data across the country. In recent years, satellite-based remote sensing combined with machine learning techniques has shown strong potential for estimating and forecasting air pollution in regions with sparse monitoring networks.

1.1 Background

Air pollution refers to the presence of harmful agents like pollutants, in the air, indoors or outdoors, that alter the natural composition of the atmosphere (WHO 2023) [1]. These pollutants can be gases, such as ozone or nitrogen oxides, or small particles like soot and dust.

Air pollution is the number one risk factor for death and disability in Nepal, surpassing malnutrition (second) and tobacco (third). It reduces life expectancy by 3.4 years for the average Nepali and causes approximately 26,000 premature deaths annually, contributing to various cardiopulmonary diseases, neonatal issues, and fertility effects [2]. The key pollutant of concern for human health is Particulate Matter (PM), which is smaller than 2.5 microns (hence named $\text{PM}_{2.5}$); due to its small size, it can penetrate deeply into the lungs and other vital organs [3].

Pollution levels are highest in the Kathmandu Valley and the Terai region. Contributing factors include industrial emissions, vehicle exhaust, cooking methods, agricultural fires, forest fires, and seasonal patterns. The transboundary pollution also significantly contributes to local air quality, with a substantial portion of $\text{PM}_{2.5}$ originating outside Nepal. The bowl-shaped topography of the Kathmandu Valley exacerbates the accumulation of pollution, preventing its dispersion. Terai experiences temperature inversions that trap air pollutants near the ground. Without effective intervention, $\text{PM}_{2.5}$ concentrations are projected to exceed the World Health Organization's guidelines. According to WHO, the acceptable 24-hour mean concentrations for $\text{PM}_{2.5}$ are $15 \mu\text{g}/\text{m}^3$, and the exceedance should occur no more than 3–4 days per year to minimize public health risks. Furthermore, WHO aims for the ultimate target for annual mean $\text{PM}_{2.5}$ concentration to be $5 \mu\text{g}/\text{m}^3$, with an initial interim goal of $35 \mu\text{g}/\text{m}^3$ to encourage incremental progress by 2035 [2][4].

Reportedly, there are 30 ground-based air quality monitoring stations in Nepal, out of which only 10 are operational [5]. In addition, ICIMOD’s 2016 report underscored irregular reporting patterns and data inconsistencies that continue to challenge reliable, continuous air quality monitoring nationwide [6]. These persistent data gaps highlight the room for improvement and the need for better, scalable, alternative approaches. In this context, satellite-based remote sensing combined with machine learning has emerged as a powerful solution to estimate $\text{PM}_{2.5}$ concentrations more comprehensively across complex and under-monitored regions. The increasing use of ML and DL in geospatial environmental remote sensing (GERS) tasks has already demonstrated significant improvements in accuracy, efficiency, and scalability — particularly for interpreting air pollution, land features, and other atmospheric elements [7].

1.2 Problem statements

Nepal currently operates only about 30 ground-based $\text{PM}_{2.5}$ monitoring stations across the country, which is far from sufficient to capture the spatial variability of air pollution in diverse regions [8]. Maintaining and expanding this network requires substantial financial investment, with the Government of Nepal bearing significant costs (Rupees 10-20 Million per monitoring station) alongside foreign aid support. However, despite this investment, these ground sensors often suffer from operational challenges, including inconsistent data reporting, frequent downtime, and maintenance issues, which severely limit the reliability and continuity of air quality information [9]. Consequently, Nepal lacks comprehensive, real-time $\text{PM}_{2.5}$ monitoring needed to inform public health and environmental policies effectively. This leaves a major research gap: the potential of using multiple satellite variables (e.g., NO_2 , SO_2 , HCHO, temperature, wind, AOD, NDVI) with machine learning models to directly predict $\text{PM}_{2.5}$ levels across Nepal’s full geography remains largely unexplored. Building such a system could enable daily, near-real-time $\text{PM}_{2.5}$ maps for both urban and rural areas, reducing reliance on expensive and sometimes faulty ground infrastructure.

1.3 Objectives

- To develop a near-real-time $\text{PM}_{2.5}$ estimation and forecasting system using satellite data.
- To compare model performance across four geographic terrains using ground truth data.

1.4 Scope / Significance of Project

- Provides cost-effective and nationwide PM2.5 monitoring using satellite data instead of expensive ground stations.
- Helps public health and policy makers by giving accurate real-time air quality maps and forecasts.
- Creates a scalable Geo-AI model that can be used in other hilly or plain areas with similar pollution issues.

2. Literature Review

2.1 Related work

This literature review aims to investigate existing research on satellite-based estimation of $\text{PM}_{2.5}$ concentrations, with a focus on the use of AOD data from sensors like MODIS and Himawari-8. By examining studies conducted in various geographical regions, specially in Asia, the objective is to understand the methodologies applied, challenges encountered—such as data gaps, seasonal variability, and cloud interference—and the modeling strategies used to address these issues. This will provide insights into how different techniques and data sources can be leveraged to improve air pollution estimation using remote sensing.

Zahra Amiri et al. focused their study on Tehran, Iran, where they modeled $\text{PM}_{2.5}$ concentrations over a ten-year period (2014–2023) using MODIS-AOD satellite data and meteorological variables. They used the Inverse Distance Weighted (IDW) method for spatial interpolation and applied machine learning models such as Random Forest (RF), Support Vector Machine (SVM), and Gaussian Process Regression (GPR). The RF model achieved the best prediction accuracy (94–98%). Notably, including meteorological data boosted the AOD– $\text{PM}_{2.5}$ correlation from 80.4% to 83% [10].

In 2021, a team of Korean researchers led by Sang-Min Kim et al. conducted a study in Seoul to estimate $\text{PM}_{2.5}$ concentrations using various AOD datasets (satellite- and ground-based), meteorological data, and multiple linear regression (MLR). They found that MODIS AOD performed better than other satellite sources, and the models achieved strong correlation ($r \geq 0.8$). Interestingly, results varied by season—with autumn showing the highest accuracy, while winter results were comparatively weaker due to cloud interference in AOD retrieval [11].

A 2024 study by Youjeong Youn et al. addressed the issue of data gaps in Himawari-8 AOD products caused by cloud coverage. Conducted over the Korean Peninsula, the study introduced a spatial gap-filling technique using a Random Forest model trained on meteorological and model-based AOD inputs (CAMS and MERRA-2). This improved the data coverage from 6% to a full 100%, with blind test performance showing RMSE of 0.064 and $R = 0.966$. Though raw satellite data were slightly more accurate, the filled dataset provided far more usable data points ($n = 4149$ vs. 346) [12].

Another study from China by Baolei Lv et al., conducted in 2016, focused on the Beijing–Tianjin–Hebei (BTH) region. The authors used a Bayesian-based downscaling algorithm to estimate daily $\text{PM}_{2.5}$ concentrations at 4 km resolution, integrating MODIS-AOD with ground observations. They also addressed missing AOD values through a two-step imputation method. The model delivered good results, with $R^2 = 0.75$ in fitting, 0.58 in random cross-validation, and 0.47 in city-specific tests [13].

In 2022, researchers led by Xinyu Yu et al. explored the Yangtze River Delta (YRD) in eastern China to overcome limitations of conventional satellite-based $\text{PM}_{2.5}$ estimation methods. They developed a novel Hierarchical Geospatial LSTM (HG-LSTM) model trained on multi-source AOD and ground sensor data to produce hourly $\text{PM}_{2.5}$ estimates at 2-km resolution. The model achieved high spatial consistency with ground truth, recording $R^2 = 0.88$ in site-based cross-validation, with over 80% of predictions deviating less than $10 \mu\text{g}/\text{m}^3$ [14].

George Kibirige et al. conducted a 2021 study in northern Taiwan where they introduced a composite neural network model (RTP) designed to capture remote transportation of pollutants. The model combined local pollution data with satellite-based remote data, dividing its architecture into two parts: STRI_{fe} for spatio-temporal feature extraction from remote areas, and STRI_p for final prediction. The addition of remote pollution signals improved the prediction accuracy, highlighting the transboundary nature of $\text{PM}_{2.5}$ transport in the region [15].

Across all reviewed studies, several recurring challenges were observed, including the difficulty of retrieving AOD data during cloudy or winter conditions, limitations in spatial and temporal resolution, and the need for reliable gap-filling techniques. While advanced machine learning models and hybrid approaches improved accuracy, issues like data sparsity and transboundary pollution remain complex. Future research should focus on integrating multi-source data, enhancing prediction under missing or noisy conditions, and adapting models to region-specific factors—especially in underrepresented areas like Nepal with diverse geography and limited ground monitoring infrastructure.

3. Related Theory

3.1 Independent Input Variables

Aerosols and AOD

Aerosols are the tiny particles like dust, smoke, or pollution which are suspended in the atmosphere, generated from natural or human causes [16]. Aerosol Optical Depth (AOD) is a measure of how much sunlight is blocked by these types of particles, and it is retrieved from the MODIS MAIAC satellite. It is considered the primary proxy for estimating PM_{2.5} concentration from satellite data, along with other variables to increase accuracy.

PM_{2.5} Relation:

- AOD tracks PM_{2.5} really well, with Pearson’s correlations around 0.7–0.85 in humid or low boundary layer conditions, especially in polluted cities during winter.
- Near-ground aerosols (below 500 m) drive the tightest AOD-PM_{2.5} link, with correlations often above 0.75, way stronger than full-column AOD [17].

Nitrogen Dioxide (NO₂)

PM_{2.5} Relation:

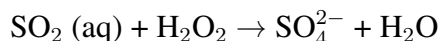
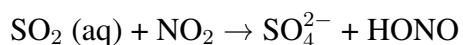
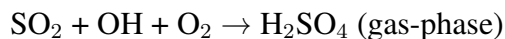
- PM_{2.5} and NO₂ often follow a similar kind of seasonal patterns, such as peaking in winter and dipping in summer, with Spearman correlations around 0.75 annually [18].
- NO₂ contributes to PM_{2.5} through oxidation of SO₂ and formation of nitrate aerosols [19][20]:
$$\text{NO}_2 + \text{HSO}_3^- \rightarrow \text{NO} + \text{H}^+ + \text{SO}_4^{2-}$$
$$\text{NO}_2 + \text{OH} \rightarrow \text{HNO}_3$$
$$\text{NO}_2 + \text{H}_2\text{O} \rightarrow \text{HONO} + \text{HNO}_3 \text{ (aerosol precursor)}$$

Sulfur Dioxide (SO₂)

PM_{2.5} Relation:

- SO₂ is a precursor for sulfate aerosols, a major component of PM_{2.5}, especially during secondary aerosol formation in humid and polluted air [21].

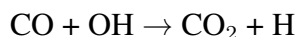
- SO₂ forms sulfuric acid via oxidation by OH, NO₂, or H₂O₂ under atmospheric conditions:



Carbon Monoxide (CO)

PM_{2.5} Relation:

- CO is a combustion tracer and correlates with PM_{2.5} from traffic and biomass burning sources.
- Though not a direct precursor, CO reacts with OH radicals and affects PM_{2.5} formation by altering oxidant balance [22]:



(Less OH → slower oxidation of SO₂/NO₂)

Formaldehyde (HCHO)

PM_{2.5} Relation:

- HCHO is a volatile organic compound that leads to secondary organic aerosol (SOA) formation, which is a major PM_{2.5} component [23].
- Through photochemistry, HCHO produces radicals and intermediates that form SOAs:

$$\text{HCHO} + \text{OH} \rightarrow \text{HO}_2 + \text{CO} \rightarrow \text{Peroxyacyl radicals} \rightarrow \text{SOA formation}$$

$$\text{HCHO} + \text{sunlight} \rightarrow \text{H} + \text{HCO} \rightarrow \text{chain propagation}$$

Ozone (O₃)

PM_{2.5} Relation:

- Ozone shows a negative or weak correlation with PM_{2.5} in colder seasons but may align positively in warmer months due to photochemical smog and oxidation of VOCs [24].
- O₃ plays a role in PM_{2.5} via oxidation pathways involving VOCs and SO₂ [25]:

$$\text{O}_3 + \text{VOC} \rightarrow \text{RO}_2 \rightarrow \text{SOA (organic PM}_{2.5}\text{)}$$

$$\text{O}_3 + \text{SO}_2 \text{ (in water)} \rightarrow \text{H}_2\text{SO}_4$$

Normalized Difference Vegetation Index (NDVI)

NDVI is a commonly used satellite-based index that measures how healthy and green vegetation is by comparing how much near-infrared (NIR) and red light the plants reflect [26].

PM_{2.5} Relation:

- NDVI often exhibits a negative correlation with PM_{2.5} concentrations, especially in vegetated or rural areas, due to the role of vegetation in air purification [24].
- It helps to capture spatial heterogeneity in surface properties that influence pollutant dispersion and deposition [27].

Surface Reflectance (SR)

Surface reflectance refers to the fraction of incoming solar radiation that is reflected by the Earth's surface, corrected for atmospheric conditions, and is used to derive many land surface products [28].

PM_{2.5} Relation:

- Surface reflectance bands, especially in visible and near-infrared, are used to estimate ground-level visibility and haze, which correlate with PM_{2.5} levels.
- These values enhance land-atmosphere interaction features in satellite-based pollution models [29].

Meteorological Variables

Meteorological conditions play a crucial role in the formation, accumulation, and dispersion of particulate matter like PM_{2.5}. Variations in temperature, humidity, wind, and precipitation influence both the chemical reactions that generate secondary particles and the physical transport or removal of aerosols from the atmosphere. The following are the meteorological parameters used in this study and their general relationship with PM_{2.5} concentrations, as summarized in Zhang et al. (2015) [30].

- **Temperature (TEMP):** Higher temperatures often enhance photochemical activity, leading to increased secondary aerosol formation and potentially elevated PM_{2.5} levels.
- **Dew Point Temperature (DPT):** Elevated dew point temperatures indicate more moisture in the air, which facilitates particle growth and hygroscopic transformation, potentially raising PM_{2.5} concentrations.
- **Wind Speed (WS):** Strong winds typically disperse particulate matter and reduce PM_{2.5} accumulation, whereas low wind speeds can contribute to pollutant buildup.

- **Wind Direction (WD):** This affects the transport of aerosols; winds from polluted regions may increase PM_{2.5}, while those from cleaner areas may dilute it.
- **Surface Pressure (SP):** High surface pressure is often linked to stable atmospheric conditions that trap pollutants near the ground, thus increasing PM_{2.5} levels.
- **Total Precipitation (TP):** Precipitation has a cleansing effect on the atmosphere by removing aerosols through wet deposition, generally leading to decreased PM_{2.5} concentrations.

3.2 Prediction Algorithms

Based on satellite data, we are planning to explore various machine learning and deep learning algorithms to estimate as well as predict PM_{2.5} concentrations. The following models can be used in our study:

3.2.1 Random Forest (RF)

Random Forest (RF) is an ensemble machine learning algorithm that builds multiple decision trees and combines their predictions. For regression tasks like estimating daily PM_{2.5} levels from satellite and meteorological features, it averages the outputs of the trees to provide a stable and accurate estimate. This reduces overfitting and handles non-linear relationships well, making it suitable for real-time air quality estimation using time-series data.

Working:

- Randomly samples data and features to construct each decision tree.
- Trains a collection of trees independently to form the “forest.”
- Aggregates predictions by averaging for regression, improving overall robustness.

Formula (Regression):

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(x) \quad \dots \quad (i)$$

Where:

- $\hat{y}$ = Predicted PM_{2.5} value
- $T_i(x)$ = Prediction from the i^{th} decision tree
- N = Number of trees in the forest
- x = Input features (e.g., AOD, temperature, etc.)

3.2.2 Extreme Gradient Boosting (XGBoost)

XGBoost is a gradient boosting algorithm that sequentially builds decision trees, with each new tree correcting errors from the previous ones. It's efficient for regression on time-series data, like estimating current PM2.5 from features such as AOD and weather variables, by minimizing loss through gradient descent. This makes it powerful for real-time estimation in diverse terrains.

Working:

- Initializes with a base model and iteratively adds trees to reduce residuals.
- Uses regularization to prevent overfitting and handles missing values natively.
- Optimizes with parallel processing for faster training on large datasets.

Formula:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad \dots \quad (\text{ii})$$

Where:

- $\hat{y}_i$ = Predicted PM2.5 for sample i
- K = Number of trees
- f_k = Function representing the k^{th} tree
- x_i = Input features for sample i

3.2.3 Light Gradient Boosting Machine (LightGBM)

LightGBM is a high-performance gradient boosting framework that grows trees leaf-wise for efficiency. It's ideal for regressing PM2.5 estimates from time-series features like satellite data, offering faster training and lower memory usage compared to level-wise methods, while maintaining high accuracy for real-time applications.

Working:

- Builds trees by selecting the leaf with maximum gain for splitting.
- Uses histogram-based algorithms to speed up feature binning.
- Handles categorical features directly and supports large-scale data.

Formula:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad \dots \quad (\text{iii})$$

Where:

- $\hat{y}_i$ = Predicted PM2.5 for sample i
- K = Number of trees
- f_k = Function representing the k^{th} tree
- x_i = Input features for sample i

3.2.4 Categorical Boosting (CatBoost)

CatBoost is a gradient boosting algorithm optimized for categorical data, using ordered boosting to reduce overfitting. For PM2.5 estimation, it effectively processes mixed features from satellites and meteo data in a time-series context, providing robust real-time predictions without extensive preprocessing.

Working:

- Encodes categorical features automatically during training.
- Applies random permutations to avoid prediction shift in boosting.
- Focuses on minimizing loss sequentially while handling imbalances.

Formula:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad \dots \quad (\text{iv})$$

Where:

- $\hat{y}_i$ = Predicted PM2.5 for sample i
- K = Number of trees
- f_k = Function representing the k^{th} tree
- x_i = Input features for sample i

3.2.5 Long Short-Term Memory (LSTM)

LSTM is a type of recurrent neural network designed for sequence data, excelling in capturing long-term dependencies. In this project, it's used for forecasting: taking the last 30 days of PM2.5 and features as input to predict the next 7 days' average PM2.5 levels, enabling week-ahead air quality predictions.

Working:

- Maintains information through cell states and gates (forget, input, output).

- Processes sequences step-by-step, updating memory to retain relevant patterns.
- Mitigates vanishing gradients, making it suitable for time-series forecasting.

Formulae:

$$\begin{aligned}
f_t &= \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \\
i_t &= \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \\
\tilde{C}_t &= \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad \dots \quad (\text{v}) \\
C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \\
o_t &= \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \\
h_t &= o_t \odot \tanh(C_t)
\end{aligned}$$

Where:

- x_t = Input sequence at time t (e.g., past PM2.5 features)
- h_t = Hidden state at time t
- C_t = Cell state at time t
- f_t, i_t, o_t = Forget, input, and output gates
- $\tilde{C}_t$ = Candidate cell state
- σ = Sigmoid function
- $\tanh$ = Hyperbolic tangent
- W, b = Weights and biases
- $\odot$ = Element-wise multiplication

3.3 Methodological Concepts

This section presents the theoretical foundations related to model development, evaluation strategies, and generalization techniques commonly applied in machine learning-based air quality prediction systems.

3.3.1 Feature Engineering

Feature engineering refers to the process of transforming raw input data into informative representations that enhance model performance. In environmental modeling, this includes temporal encoding, derived meteorological variables, dimensionality reduction, and statistical transformations.

Seasonal periodicity is often represented using cyclic encoding to preserve the continuity of time. Since the day-of-year (DOY) variable is cyclical, sine and cosine transformations are applied:

$$\text{DOY}_{\sin} = \sin\left(\frac{2\pi \cdot \text{DOY}}{365}\right), \quad \text{DOY}_{\cos} = \cos\left(\frac{2\pi \cdot \text{DOY}}{365}\right) \quad \dots \quad (\text{vi})$$

This transformation avoids artificial discontinuity between December 31 and January 1 and allows models to capture seasonal trends effectively.

Dimensionality reduction techniques, such as feature importance ranking from tree-based models, are commonly applied to identify the most informative spectral bands or predictors. This improves computational efficiency and reduces overfitting.

3.3.2 Data Splitting Strategies

Data splitting is a fundamental component of model evaluation. It determines how data are partitioned into training and testing subsets and directly influences performance assessment.

Random Split

Random splitting divides the dataset into training and testing subsets without considering temporal order. Each observation has an equal probability of being assigned to either subset. This method evaluates the model's ability to generalize across the full distribution of the dataset and is commonly applied when temporal dependency is not the primary focus.

Chronological Split

Chronological splitting preserves temporal order by using earlier observations for training and later observations for testing. This approach is particularly important in time-series forecasting tasks, as it prevents information leakage from future data and provides a realistic assessment of operational performance.

3.3.3 Transfer Learning

Transfer learning is a machine learning technique in which knowledge gained from one domain or dataset is applied to improve performance in another related domain. Instead of training a model from scratch, learned patterns from a source dataset are transferred to a target dataset.

In environmental applications, transfer learning is especially useful when data availability is limited in certain locations. Shared atmospheric processes and physical relationships between pollutants allow models trained in one region to generalize to others, provided that domain differences are properly managed.

3.3.4 Ensemble Learning and Stacking

Ensemble learning combines multiple predictive models to improve overall accuracy and robustness. The underlying principle is that aggregating diverse learners reduces variance, bias, or both.

Tree-based ensemble methods such as Random Forest and Gradient Boosting construct multiple decision trees and aggregate their outputs to produce a final prediction.

Stacking (stacked generalization) is an advanced ensemble technique in which predictions from multiple base models are used as inputs to a higher-level meta-learner:

$$\hat{y} = g(f_1(x), f_2(x), \dots, f_k(x)) \quad \dots \quad (\text{vii})$$

where $f_1, f_2, \dots, f_k$ represent base learners and g represents the meta-model.

Stacking allows the meta-learner to identify patterns in the strengths and weaknesses of individual models, often leading to improved predictive performance compared to single-model approaches.

4. Methodology

4.1 Requirement Analysis

4.1.1 Functional Requirements

R.1: Region Selection

Description: The Region Selection function allows the user to select a geographical region to retrieve air quality data. The system presents an interactive user interface where the user can choose a region from a dropdown menu and initiate data retrieval by clicking a "Get Data" button.

- **R.1.1: Display Region Selection Interface**

Input: User accesses the platform via web or mobile application.

Output: Interactive UI with a dropdown menu listing available regions.

- **R.1.2: Select Region**

Input: User selects a region from the dropdown menu.

Output: Selected region is highlighted, enabling the "Get Data" button.

- **R.1.3: Initiate Data Retrieval**

Input: User clicks the "Get Data" button.

Output: System triggers backend data fetch for the selected region.

Processing: The system validates the region selection and sends a request to the backend API.

R.2: Backend Data Fetch

Description: The Backend Data Fetch function queries real-time air quality data for the user-selected region via an API.

- **R.2.1: Query Real-Time Data**

Input: Selected region from R.1.3.

Output: Real-time air quality data retrieved from the API in JSON format.

Processing: The backend sends an API request and receives structured data containing PM_{2.5} levels and related metrics.

R.3: Model Prediction

Description: The Model Prediction function processes the retrieved data using machine learning (ML) and deep learning (DL) algorithms to predict PM_{2.5} levels and trends.

- **R.3.1: Process Data with ML/DL Algorithms**

Input: Real-time air quality data from R.2.1.

Output: Predicted PM_{2.5} values and trend analysis.

Processing: The system applies trained ML/DL models to analyze data and generate predictions.

- **R.3.2: Package Data into Structured Response**

Input: Processed data and predictions from R.3.1.

Output: Structured JSON response containing PM_{2.5} levels, predictions, and trends.

Processing: The system formats the data for frontend consumption.

R.4: Display of Information

Description: The Display Information function presents the air quality data and predictions on the frontend, including visualizations of PM_{2.5} levels and trends.

- **R.4.1: Receive Structured Data**

Input: JSON response from R.3.2.

Output: Data parsed and ready for visualization.

- **R.4.2: Visualize Feature Trends**

Input: Parsed data from R.4.1.

Output: Graphs displaying trends of features used in ML/DL models.

Processing: The frontend renders line or bar charts to show historical and predicted trends.

- **R.4.3: Display PM_{2.5} Levels**

Input: Current PM_{2.5} data from R.4.1.

Output: PM_{2.5} level displayed with a colored background (green for healthy, orange for moderate, red for unhealthy) based on standard guidelines.

- **R.4.4: Display PM_{2.5} Trend**

Input: Trend data from R.4.1.

Output: Graph showing the historical trend of PM_{2.5} levels.

- **R.4.5: Display Predicted PM_{2.5} Value and Trend**

Input: Predicted data from R.4.1.

Output: Predicted PM_{2.5} value and trend graph (indicating increasing or decreasing).

R.5: Data Storage

Description: The Data Storage function saves calculated and predicted PM_{2.5} values in a database

for future reference and visualization.

- **R.5.1: Store Calculated and Predicted Data**

Input: Structured data from R.3.2.

Output: Data stored in the database.

Processing: The system inserts or updates records in a database with consistent data types and formats.

4.1.2 Non-Functional Requirements

R.1: Performance Requirements

- **R.1.1:** The website shall load within 3 seconds under normal network conditions.
- **R.1.2:** API responses shall be returned within 2 seconds for real-time data queries.
- **R.1.3:** The system shall handle high traffic without degradation, supporting concurrent users effectively.

R.2: Availability Requirements

- **R.2.1:** The system shall maintain an uptime of at least 90%.
- **R.2.2:** The system shall be available 24/7 for user access.

R.3: Usability Requirements

- **R.3.1:** The interface shall be user-friendly, clean, and intuitive for both technical and non-technical users.
- **R.3.2:** The PM_{2.5} index shall adhere to standard air quality guidelines for healthy and harmful levels.

R.4: Compatibility Requirements

- **R.4.1:** The platform shall operate seamlessly across modern browsers (Chrome, Firefox, Edge, Safari) and mobile devices (Android and iOS).
- **R.4.2:** The backend shall be developed using Python, ensuring compatibility with ML/DL frameworks.

R.5: Data Consistency Requirements

- **R.5.1:** All stored data shall maintain consistent data types and formats across the database.
- **R.5.2:** Data visualizations shall reflect consistent units and scales for PM_{2.5} measurements.

4.2 Data Collection

Data were collected from 12 sites across Nepal and India, selected based on the availability of ground-based $PM_{2.5}$ measurements from OpenAQ. Eight of these served as training locations: five within the Kathmandu Valley in Nepal (site_a through site_e) and three near the Nepal–India border in India (india_a through india_c). The remaining four sites in Nepal (Kathmandu, Pokhara, Chitwan, and Birgunj) served as deployment locations, where the goal is real-time $PM_{2.5}$ prediction. Daily average $PM_{2.5}$ values from all 12 sites served as ground truth for model training, validation, and evaluation. Satellite data from MODIS, Sentinel-2, and Sentinel-5P, along with meteorological variables from ERA5, were extracted at each sensor’s coordinates to capture spectral, atmospheric, and environmental conditions relevant to $PM_{2.5}$. Table 4.1 lists the sources and key parameters used.

Table 4.1: Datasets Used

Source (Platform)	Key Parameters
ERA5 Reanalysis (Google Earth Engine)	Temperature, Relative Humidity, Dew Point Temperature, Precipitation, Surface Pressure, Wind Speed, Wind Direction, Surface Water Runoff, Soil Water Content, Boundary Layer Height, Solar Radiation, Thermal Radiation, Leaf Area Index (High/Low Vegetation), Evaporation
MODIS (Google Earth Engine)	Aerosol Optical Depth at 550 nm (AOD550)
Sentinel-2 (Google Earth Engine)	Bands B1–B9, B8A, B11, B12; Aerosol Optical Thickness (AOT), Water Vapor (WVP)
Sentinel-5P (TROPOMI) (Google Earth Engine)	NO_2 , SO_2 , HCHO, O_3 , CO, Aerosol Index
OpenAQ API (Ground Truth for Training/Validation)	Daily Average $PM_{2.5}$

4.2.1 ERA5 Reanalysis Data

Meteorological and land surface variables were collected from the ERA5 reanalysis product. The retrieved parameters include air temperature, dew point temperature, relative humidity, precipitation, surface pressure, wind speed, wind direction, surface water runoff, soil water content, evaporation, solar radiation, thermal radiation, leaf area indices, and boundary layer height.

Some variables, such as relative humidity and wind characteristics, were derived from primary ERA5 parameters to ensure physical consistency. All ERA5 variables were aggregated to daily temporal resolution and spatially averaged within an approximate 5.5 km radius buffer to match the spatial scale of satellite-derived datasets. ERA5 coverage begins from January 2000, providing long-term temporal continuity across all satellite missions used in this study.

4.2.2 MODIS Aerosol Data

Aerosol Optical Depth (AOD) at 550 nm was collected from MODIS products to represent columnar aerosol loading, which is strongly associated with surface-level $\text{PM}_{2.5}$. MODIS data were processed at daily temporal resolution and spatially averaged using circular buffers of 5000 m and 600 m radius.

Quality assurance flags were applied to remove low-confidence retrievals and cloud-contaminated pixels before aggregation. The processed MODIS AOD data are available starting from February 2000 and serve as a key proxy linking satellite observations to ground-level particulate pollution.

4.2.3 Sentinel-2 Optical and Atmospheric Data

Sentinel-2 data were collected to capture high-resolution optical and atmospheric information related to surface and near-surface conditions. Multiple spectral bands spanning the visible, near-infrared, and shortwave infrared regions were retrieved, along with atmospheric products such as Aerosol Optical Thickness and Water Vapor Product.

All Sentinel-2 data were aggregated to daily resolution using fixed spatial buffers of 5000 m and 600 m radius. Cloud masking and quality filtering were applied before aggregation to reduce noise. Sentinel-2 data are available from 2017 onward.

4.2.4 Sentinel-5P Trace Gas Observations

Trace gas concentrations and absorbing aerosol indices were obtained from Sentinel-5P to represent atmospheric composition linked to emission sources and secondary aerosol formation. The extracted variables include nitrogen dioxide, sulfur dioxide, carbon monoxide, ozone, formaldehyde, and aerosol index.

These variables were processed to daily aggregates and spatially averaged within circular buffers of 5000 m and 600 m radius, consistent with the other satellite sources. Sentinel-5P coverage is available from mid-2018.

4.2.5 Ground-Based PM_{2.5} Measurements

Daily average PM_{2.5} concentrations were collected from the OpenAQ platform to serve as ground truth observations. These measurements were obtained from regulatory and low-cost monitoring stations and represent point-based sensor data.

PM_{2.5} data for the eight training sites provided extended historical records used for model development, while data from the four deployment sites were used for evaluation and real-time prediction. The OpenAQ PM_{2.5} values were temporally aligned with satellite and reanalysis datasets. Due to varying sensor availability, temporal coverage differs by location; however, all available observations were retained to maximize training and evaluation samples.

4.3 Data Pre-processing

ERA5 reanalysis data was complete across all sites and dates, requiring no gap-filling. However, the three satellite sources had substantial gaps due to cloud cover, sensor limitations, and revisit frequency. Since the downstream PM_{2.5} models require continuous daily inputs, the preprocessing pipeline focused on two main tasks: satellite gap-filling and temporal feature engineering and dataset alignment.

As noted earlier, the training and deployment sites have different operational requirements. Training sites only need complete historical data, whereas deployment sites must also support real-time prediction with no access to future observations. This distinction shaped the gap-filling strategy, as described below.

4.3.1 Extent of Missing Satellite Data

Table 4.2 summarizes the overall percentage of missing observations for each satellite source.

Table 4.2: Overall missing-data rates for each satellite source.

Satellite	Variables	Overall Missing %	Range across sites
MODIS AOD ₅₅₀	1 band	~40–50%	36–58%
Sentinel-2	14 bands	~53–54%	31–79%
Sentinel-5P	6 gases	~0.5–69%	25–79%

MODIS exhibited approximately 40% missing data at training sites and close to 50% at deployment sites. Sentinel-2 had the highest overall gap rate (over 50%), with some sites exceeding 75% missing observations. Sentinel-5P missingness varied widely by gas: O₃ was nearly complete (~5% missing), while SO₂ was missing over 60% of the time. These high gap rates necessitated a systematic filling strategy to produce continuous feature inputs.

4.3.2 Gap-Filling Strategy

Machine learning-based gap-filling models were trained to predict missing satellite values using ERA5 meteorological variables, temporal features such as month, day-of-year, week-of-year, season indicators, and cyclic sine/cosine encodings, and where applicable, the satellite variable’s own lagged values and rolling statistics.

Training Sites: At the eight training sites, the gap-filling models were trained using a random split of the available historical data. Since these sites do not require real-time prediction, both past and future observations could serve as context. This allowed the models to learn from the broadest possible range of conditions, maximizing the quality of the filled dataset.

Deployment Sites: At the four deployment sites, the gap-filling models were trained using a chronological split to simulate real-time conditions. Only past data was used to predict missing values, ensuring that no future information leaked into the filled features. This is essential because during live operation, satellite gaps must be filled using only data available up to the current day. Input features were therefore restricted to same-day ERA5 variables, temporal encodings, and for MODIS, backward-looking lagged values and rolling statistics of the satellite’s own past observations.

4.3.3 MODIS AOD₅₅₀ Gap-Filling

MODIS AOD₅₅₀ was filled using gradient-boosted tree models: XGBoost or LightGBM. The input feature set included all ERA5 variables, temporal encodings, AOD lagged values, rolling statistics, ERA5 rolling statistics, interaction features, and monthly AOD climatology.

For deployment sites, a transfer learning approach was also evaluated, where models trained on the completed training-site datasets were applied to deployment locations to leverage cross-site meteorological and seasonal patterns. Transfer learning improved gap-filling at three of the four deployment sites, while Pokhara performed better with a site-specific model due to its unique valley microclimate.

4.3.4 Sentinel-2 Multispectral Gap-Filling

Each of the 14 Sentinel-2 bands was filled independently using a stacking ensemble of three base learners: XGBoost, LightGBM, CatBoost with a Ridge regression meta-learner. The input features comprised ERA5 variables, temporal and cyclic encodings, season flags, band-specific climatology, and interaction features.

Unlike MODIS, the Sentinel-2 models did not use lagged satellite values as inputs. Transfer learn-

ing across sites was not applied for Sentinel-2 because high-resolution surface reflectance is highly location-specific and lacks the consistent cross-site temporal patterns that MODIS AOD exhibits.

4.3.5 Sentinel-5P Tropospheric Gas Gap-Filling

The six Sentinel-5P gas columns were each filled using a per-site, per-gas model: XGBoost, LightGBM, or CatBoost, selected based on validation performance. Input features followed the same ERA5 + temporal + climatology schema as Sentinel-2, with gas-specific monthly and weekly climatology values. Lagged satellite values were not used.

The predictability of Sentinel-5P variables varied considerably. Gases with strong meteorological control were well-suited to ERA5-based prediction, while short-lived or emission-driven species were difficult to predict from meteorological data alone. For these poorly predictable variables, the downstream $\text{PM}_{2.5}$ ensemble model handles missing inputs natively, relying only on authentic satellite observations when gap-filled values would be unreliable.

4.3.6 Temporal Feature Engineering and Dataset Alignment

To capture seasonal and cyclic variations in air pollution, sine and cosine transformations of the day of the year were computed and appended as temporal features. These encodings allow models to learn smooth periodic patterns without introducing artificial discontinuities at year boundaries.

All datasets were then synchronized to a common daily temporal resolution, merged by date and site, and combined into a single structured dataset. $\text{PM}_{2.5}$ serves as the target variable, while all satellite, reanalysis, and temporal parameters serve as input features. This unified dataset forms the foundation for the ensemble and LSTM-based models described in subsequent sections.

4.4 Model Development Approach

The **Model Development Approach** consists of three main components that together enable real-time $\text{PM}_{2.5}$ estimation and short-term forecasting at deployment sites:

1. **Near Real-Time (NRT) Synthesis Module** – generates continuous daily meteorological and satellite-derived features.
2. **Ensemble $\text{PM}_{2.5}$ Prediction Model** – predicts current-day $\text{PM}_{2.5}$ using the outputs of the NRT synthesis module and engineered features.
3. **LSTM $\text{PM}_{2.5}$ Forecasting Model** – forecasts $\text{PM}_{2.5}$ for the next seven days using historical observations and NRT features.

4.4.1 Near Real-Time (NRT) Synthesis Module

The NRT Synthesis Module provides complete, daily $\text{PM}_{2.5}$ -related features, even when satellite data are missing or delayed. It integrates meteorological and satellite-based observations with machine learning models. This module is composed of four site-specific predictors: ERA5 Forecaster, MODIS Predictor, Sentinel-2 Predictor, and Sentinel-5P Predictor. Each predictor is trained individually for the four deployment sites: Kathmandu, Pokhara, Chitwan, and Birgunj.

ERA5 Forecaster

The ERA5 Forecaster addresses the 7–8 day delay in operational ERA5 data by forecasting meteorological variables using:

- **LSTM for Continuous Variables:** Forecasts temperature, dew point, relative humidity, wind speed and direction, boundary layer height, and radiation fluxes using 45-day historical sequences.
- **XGBoost Rain Classifier:** Predicts rain occurrence; if rain is predicted, LSTM estimates precipitation amount; otherwise, precipitation is set to zero.

This hybrid approach ensures a continuous, gap-free meteorological feature set for downstream models.

MODIS Predictor

The MODIS Predictor estimates daily Aerosol Optical Depth (AOD_{550}) values at deployment sites, even when satellite measurements are missing. Inputs include:

- Historical MODIS AOD values
- ERA5 Forecaster outputs
- Climatology-based temporal features (e.g., day-of-year sine/cosine transformations)

Site-specific models ensure accurate reconstruction for each deployment location.

Sentinel-2 Predictor

The Sentinel-2 Predictor reconstructs daily values for multiple optical and atmospheric bands, including visible, near-infrared, and shortwave-infrared bands. Inputs include:

- Past Sentinel-2 band observations
- ERA5-derived meteorological features
- Site-specific historical patterns

Each spectral band is predicted independently to preserve high-resolution surface and atmospheric information.

Sentinel-5P Predictor

The Sentinel-5P Predictor provides estimates for tropospheric trace gases and aerosol indices: CO, NO₂, O₃, HCHO, SO₂, and Absorbing Aerosol Index (AI). Inputs include:

- Historical Sentinel-5P features
- Forecasted ERA5 variables
- Temporal features capturing seasonal patterns

Low-predictability features (e.g., SO₂, HCHO, AI) are allowed to remain missing; downstream PM_{2.5} models handle these gaps by relying on authentic observations.

Integration and Real-Time Deployment

The outputs from the four predictors are combined to form a complete set of daily features for each deployment site. These features serve as inputs for PM_{2.5} prediction and forecasting models, ensuring continuous real-time operation.

4.4.2 Ensemble PM_{2.5} Prediction Model

The Ensemble PM_{2.5} Prediction Model produces daily PM_{2.5} estimates at each deployment site by combining the outputs of the NRT Synthesis Module with engineered features. A stacking ensemble strategy is employed:

- **Base Learners:** XGBoost, LightGBM, CatBoost
- **Meta-Learner:** Ridge Regression, which aggregates base learner predictions
- **Stacking Procedure:** 5-fold cross-validation; predictions from base learners serve as inputs to the meta-learner

Input Features: A combination of satellite-derived and ERA5-based variables, including:

Table 4.3: Key input feature categories for the Ensemble PM_{2.5} model.

Feature Type	Description
Satellite	MODIS AOD, Sentinel-2 spectral bands, Sentinel-5P trace gases (NRT-filled)
Meteorology	ERA5 forecasted variables: temperature, humidity, wind, precipitation, radiation, boundary layer height
Temporal	Cyclical encoding: sin/cos of day-of-year, month, week; other calendar features
Derived / Engineered	Rolling statistics (3, 7, 14 days), lag features, and interaction terms (e.g., AOD $\times$ meteorology)

Training Setup: Chronological 75%/25% split across multiple Nepali and Indian sites. The target PM_{2.5} values are log-transformed during training to stabilize variance and inverse-transformed for predictions. The model captures complementary strengths of base learners: XGBoost for complex interactions, LightGBM for smoother patterns, and CatBoost for robustness to noisy features.

4.4.3 LSTM PM_{2.5} Forecasting Model

The LSTM PM_{2.5} Forecasting Model predicts daily PM_{2.5} for the next seven days using historical observations and NRT features. This model is designed to capture temporal dependencies and the influence of multi-source data.

- **Input Window:** Past 30 days of PM_{2.5} and corresponding NRT satellite and ERA5 features
- **Output Window:** Next 7 days of daily average PM_{2.5}

- **Architecture:** Bidirectional LSTM with attention mechanism to focus on relevant historical patterns
- **Sites:** Kathmandu, Pokhara, Chitwan, Birgunj
- **Train/Test Split:** Chronological 75%/25% per site

Feature Engineering:

- Temporal features: day-of-year, month, day-of-week, week-of-year, is-weekend
- Cyclical encoding: sine and cosine transforms for month, day, and week
- Rolling statistics: mean, standard deviation, max, min over 3, 7, 14-day windows
- Lag features: $PM_{2.5}$ values at lags of 1, 2, 3, 7, 14 days
- Difference features: 1-day and 7-day differences
- Exponential moving averages over 7 and 14-day windows

This configuration allows the LSTM to model both short-term fluctuations and longer-term trends in $PM_{2.5}$, integrating historical measurements with real-time satellite and meteorological data for accurate 7-day forecasting.

4.5 System Implementation Overview

The AI and machine learning components of the system were developed using Python. Core data manipulation and analysis relied on NumPy and Pandas, while visualizations were generated using Matplotlib. Machine learning models utilized scikit-learn for general utilities and specialized gradient boosting libraries, including XGBoost, LightGBM, and CatBoost, for the ensemble $PM_{2.5}$ prediction model. Deep learning architectures, specifically the LSTM-based forecasting model, were implemented in PyTorch. Hyperparameter tuning and optimization were performed using Optuna, and satellite data retrieval was handled through the earthengine-api library.

The system follows a modular development approach, with key components including:

- **Data Retrieval:** Automated access to satellite (MODIS, Sentinel-2, Sentinel-5P) and ERA5 reanalysis data.
- **Near Real-Time Synthesis Module:** Integration of ERA5 forecasts with satellite-derived predictors to generate complete environmental features for each deployment site.
- **Ensemble $PM_{2.5}$ Prediction:** Combines multiple gradient boosting learners in a stacking framework to estimate site-specific daily $PM_{2.5}$ concentrations.

- **LSTM PM_{2.5} Forecasting:** Uses historical PM_{2.5} measurements and synthesized environmental features to forecast 7-day PM_{2.5} averages.

This implementation strategy ensures reproducibility, modularity, and ease of integration with downstream applications.

4.5.1 Model Validation and Testing Strategy

All models were evaluated using a chronological train-test split rather than random sampling, so that evaluation reflects real deployment conditions where future data is never available during training. For the ERA5 LSTM forecaster and satellite gap-filling models, earlier periods were used for training and later periods held out for testing, preserving temporal order.

The ensemble PM_{2.5} prediction model was validated against ground-truth sensor measurements at each site. For the 7-day LSTM forecaster, ground-truth PM_{2.5} records were limited, insufficient for direct LSTM training. To address this, ensemble model predictions over the historical period were used to extend the PM_{2.5} sequence, giving the LSTM a longer input history. The forecaster was then validated against actual ground-truth measurements on the held-out period, ensuring that despite augmented training, final evaluation remained grounded in real observations.

Site-level validation ensures performance is assessed in the spatial context of each model rather than on pooled or averaged data. For the LSTM forecaster, accuracy was also evaluated per forecast day across the 7-day horizon to quantify how prediction quality changes with lead time.

Model performance is reported using R^2 , Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and a MAPE-based accuracy metric (%).

5. System design

5.1 System Architecture

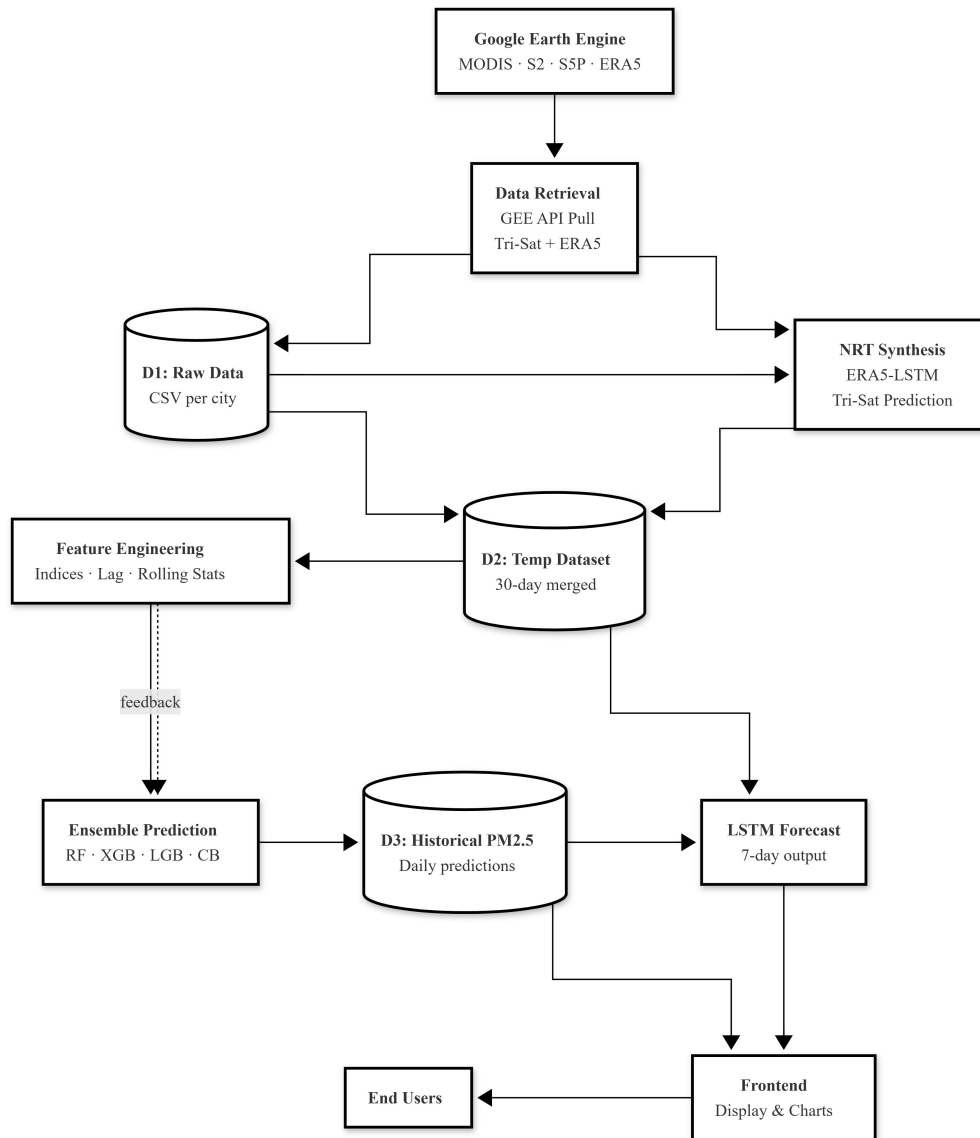


Figure 5.1: Block Diagram of the System

Figure 5.1 illustrates the system architecture, organized into three tiers: data acquisition, backend processing, and frontend dissemination. This layered design separates data retrieval, modeling, and user presentation.

5.1.1 Data Acquisition

All external data are accessed through the Google Earth Engine (GEE) platform. The system uses MODIS, Sentinel-2, and Sentinel-5P satellites for aerosol, land surface, and trace gas measurements respectively. Meteorological variables are obtained from ERA5 reanalysis data. Due to ERA5's publication delay of approximately 7–8 days, the most recent conditions are estimated using internal predictive models to ensure uninterrupted daily operation.

5.1.2 Data Storage

Three main repositories support backend operations:

- **Raw Data Store (D1):** An append-only archive of all observations from GEE, preserving missing values and temporal gaps.
- **Temporary Dataset (D2):** Rebuilt at each pipeline run, containing a harmonised 30-day rolling window of observed or model-predicted values aligned across all sources.
- **Historical PM2.5 Store (D3):** Stores daily PM2.5 predictions, supports frontend visualisation, and provides input sequences for the forecasting model.

5.1.3 Processing Pipeline

The backend processing is structured in five stages:

1. **Near Real-Time Synthesis:** Resolves data gaps using predictive modelling.
2. **Feature Engineering:** Derives structured predictors from the synthesised dataset.
3. **Ensemble PM2.5 Prediction:** Combines four tree-based models through stacking to estimate daily PM2.5 concentrations.
4. **LSTM Forecasting:** Produces seven-day PM2.5 forecasts based on historical prediction sequences.
5. **Frontend Display:** Publishes predictions and forecasts for each monitored city through a web interface.

5.1.4 Near Real-Time Synthesis and Prediction Pipeline

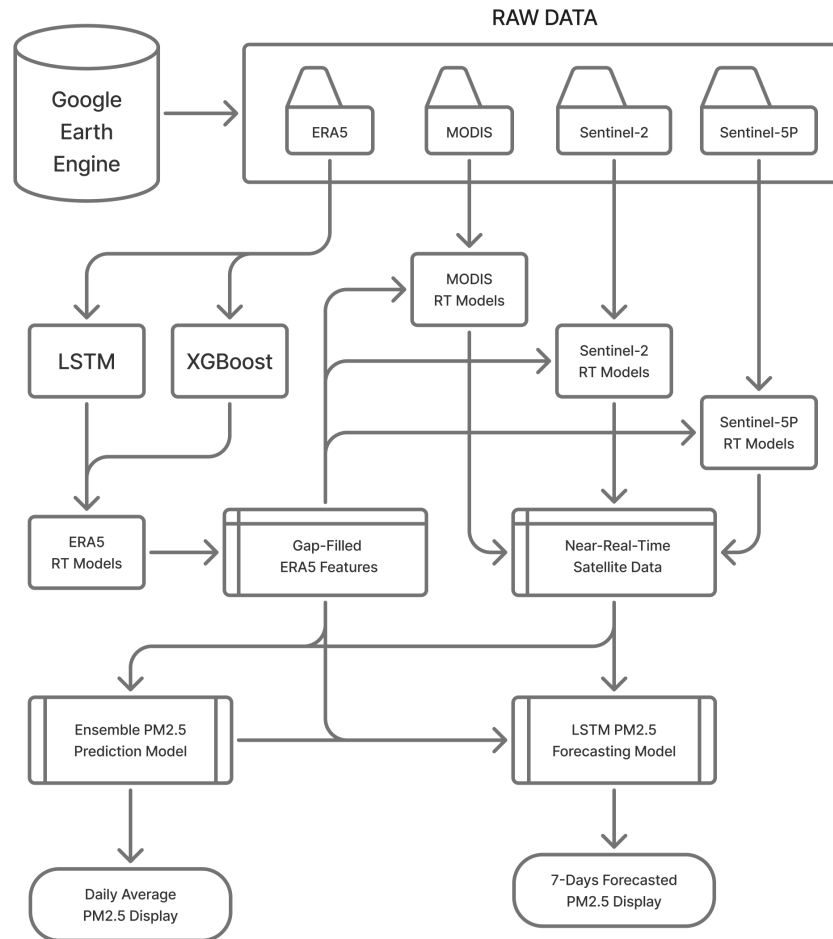


Figure 5.2: End-to-end near real-time synthesis and PM_{2.5} prediction pipeline.

Figure 5.2 illustrates the end-to-end daily pipeline. Raw data from Google Earth Engine is split into four streams: ERA5 reanalysis, MODIS AOD, Sentinel-2 surface reflectance, and Sentinel-5P trace gases. Since ERA5 carries a publication delay of approximately 7–8 days, an LSTM and an XGBoost rain classifier fill the missing ERA5 variables. Each satellite product similarly has dedicated gap-filling models driven by the available ERA5 features. The resulting complete feature set is fed into the ensemble PM_{2.5} prediction model for daily estimates, while accumulated historical predictions and synthesised features serve as input to the LSTM for 7-day ahead forecasting.

5.2 System Working Flow

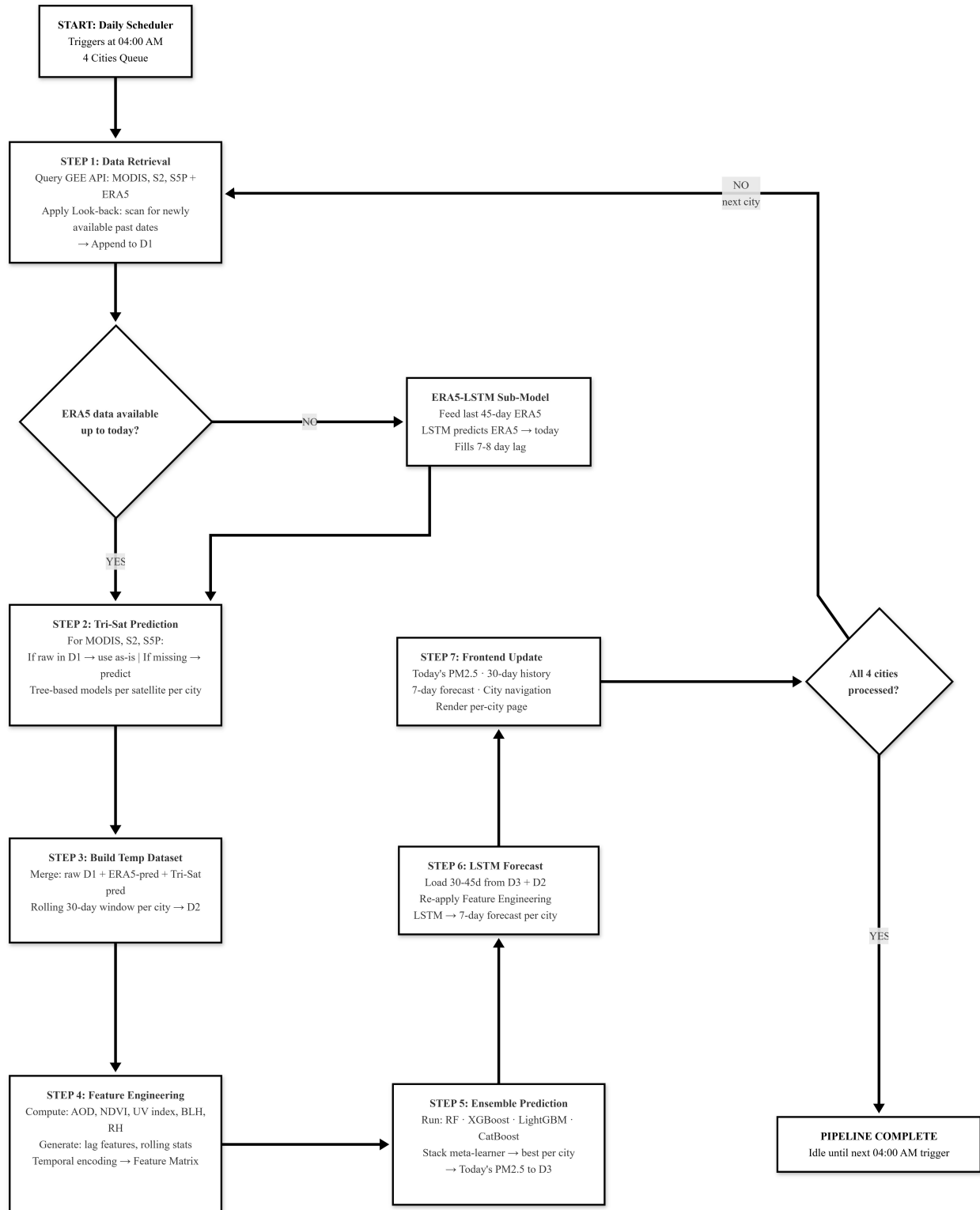


Figure 5.3: Workflow of the System

Figure 5.3 presents the operational workflow of the daily automated pipeline, which executes at 04:00 AM for four deployment cities: Kathmandu, Pokhara, Birgunj, and Chitwan. Each city is processed sequentially in a self-contained loop, allowing independent execution; a failure in one city does not affect the others.

Step 1: Data Retrieval

The pipeline queries Google Earth Engine for the latest satellite and ERA5 records within each city's bounding box. A look-back mechanism ensures previously missing or delayed records are retrieved and appended to the Raw Data Store (D1).

Step 2: Near-Real-Time Data Synthesis

This stage fills gaps in ERA5 and satellite data. Missing ERA5 values are predicted using ERA5-LSTM model, and Tri-Sat models estimate any missing satellite parameters, producing a coherent, gap-free dataset stored in D2.

Step 3: Feature Engineering

D2 is transformed into a structured feature matrix including spectral indices, AOD interactions, atmospheric proxies, meteorological interactions, temporal lags (1, 3, 7 days), rolling statistics (7, 14, 30 days), and cyclical encoding for day-of-year and season.

Step 4: Ensemble PM2.5 Prediction

The engineered features are input to the ensemble prediction model, consisting of Random Forest, XGBoost, LightGBM, and CatBoost. Outputs are combined using a stacking meta-learner to generate the daily PM2.5 concentration, which is recorded in the Historical PM2.5 Store (D3).

Step 5: LSTM Forecasting

The LSTM forecasting model generates seven-day PM2.5 predictions using the previous 30–45 days of D3 predictions and corresponding D2 features. The Feature Engineering Module is reapplied to maintain input consistency. Forecast outputs are ready for frontend publication.

Step 6: Frontend Update

The frontend layer is refreshed with the current-day prediction, seven-day forecast, and 30-day historical trends. After all four cities are processed, the pipeline idles until the next scheduled run. Temporary D2 data is discarded, while D1 and D3 are persisted.

5.3 Data Flow Diagrams

This section presents the Data Flow Diagram (DFD) of the PM_{2.5} prediction and forecasting system, illustrating the movement of data between external sources, processing modules, and output interfaces.

5.3.1 Level 0 DFD

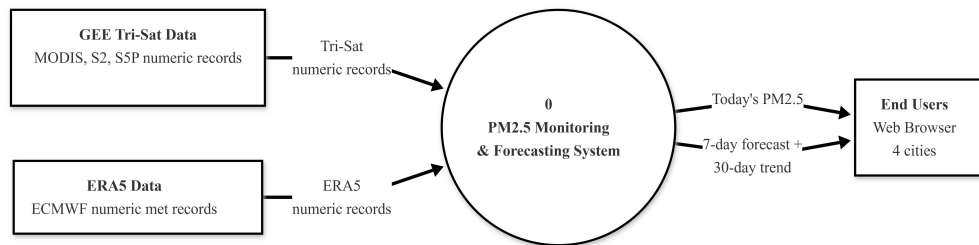


Figure 5.4: DFD - Level 0: Context Diagram

Figure 5.4 presents the Level 0 Data Flow Diagram of the PM_{2.5} monitoring and forecasting system. The system is represented as a single process interacting with three external entities: satellite data sources via Google Earth Engine, ERA5 meteorological reanalysis, and end users who access predictions and forecasts.

5.3.2 Level 1 DFD

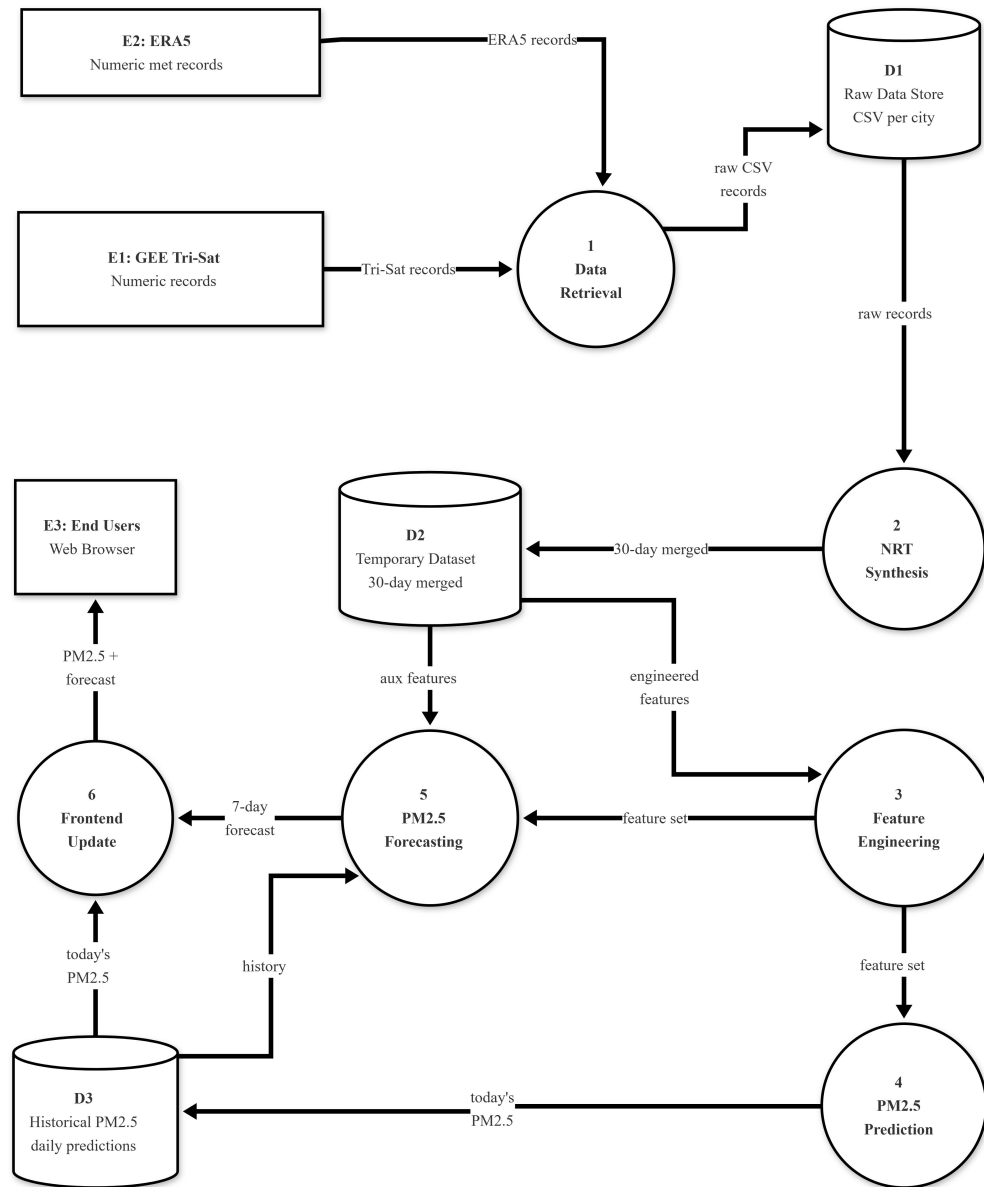


Figure 5.5: DFD - Level 1: High-Level System Processes

Figure 5.5 decomposes the system into six main processes, three data stores, and two external entities. It illustrates the flow of satellite and meteorological data through successive stages of preprocessing, feature extraction, PM_{2.5} prediction, and LSTM-based forecasting. Processed information is stored and delivered to end users via the web interface.

5.3.3 Level 2 DFDs

DFD Level 2.1: Data Retrieval Module

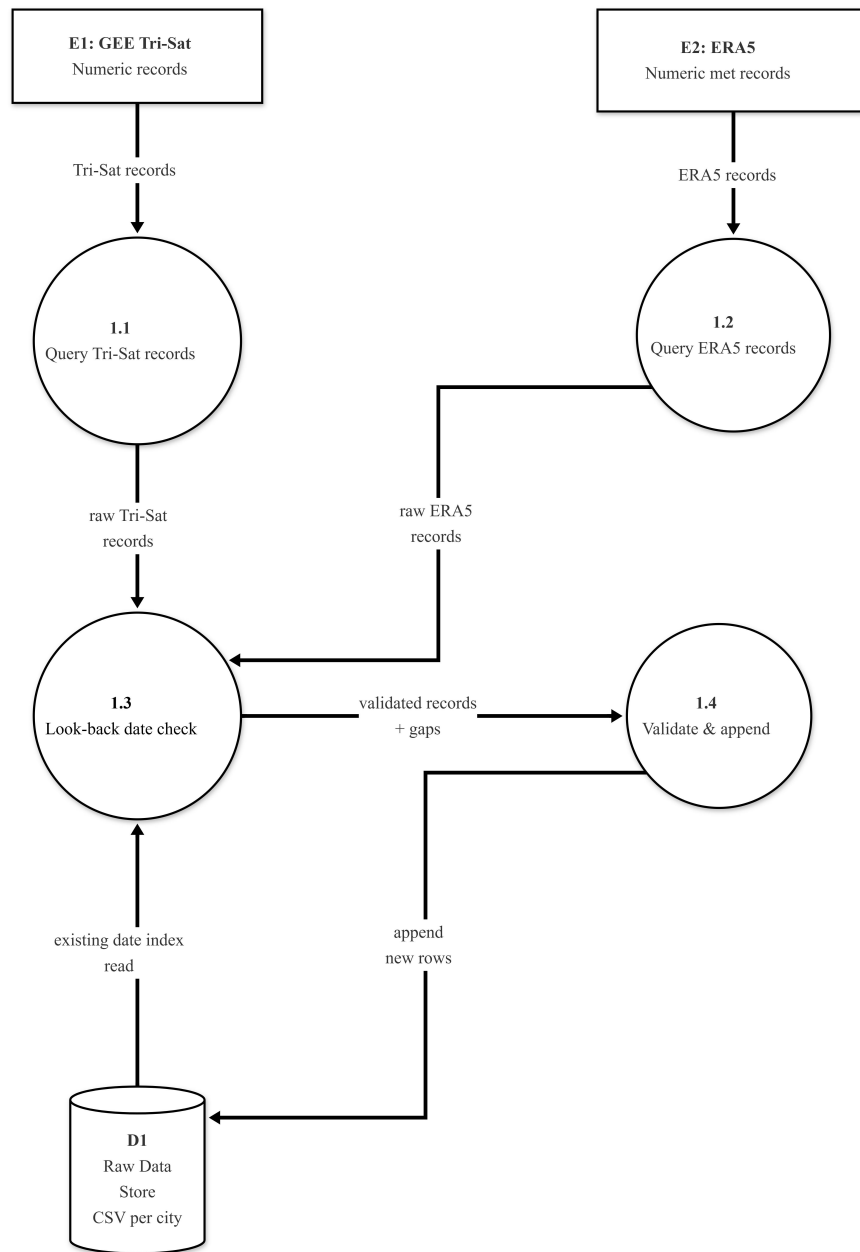


Figure 5.6: DFD - Level 2.1: Data Retrieval Module

Figure 5.6 shows the Data Retrieval Module (P1) broken into four steps: querying Tri-Sat products from GEE, retrieving ERA5 data, fetching missing historical records, and validating/appending data to D1.

DFD Level 2.2: NRT Synthesis Module

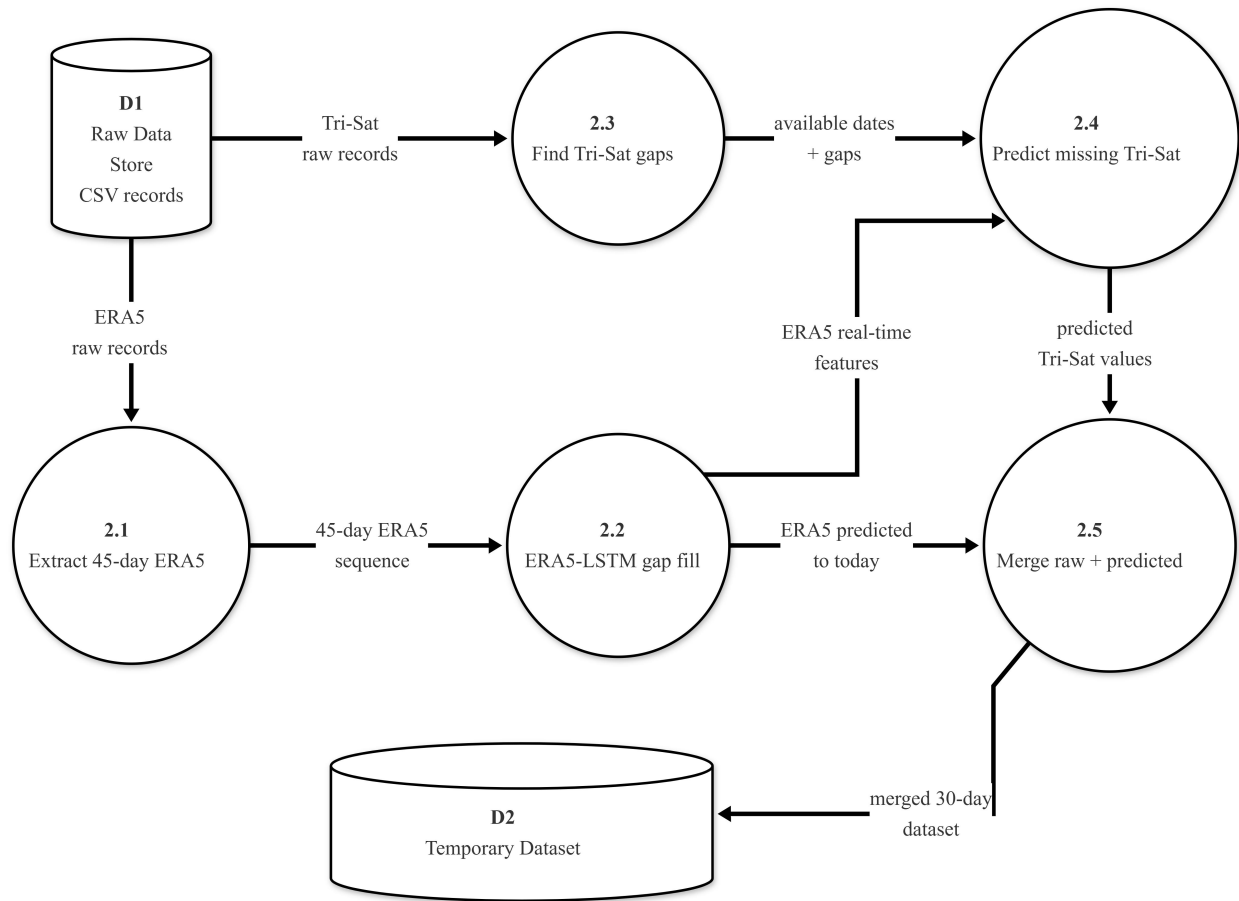


Figure 5.7: DFD - Level 2.2: Near Real-Time Synthesis Module

Figure 5.7 shows how temporal gaps in input data are filled. ERA5-LSTM predicts missing meteorological values, Tri-Sat gaps are estimated with regression models, and all data is merged into D2 for downstream processing.

DFD Level 2.3: Feature Engineering Module

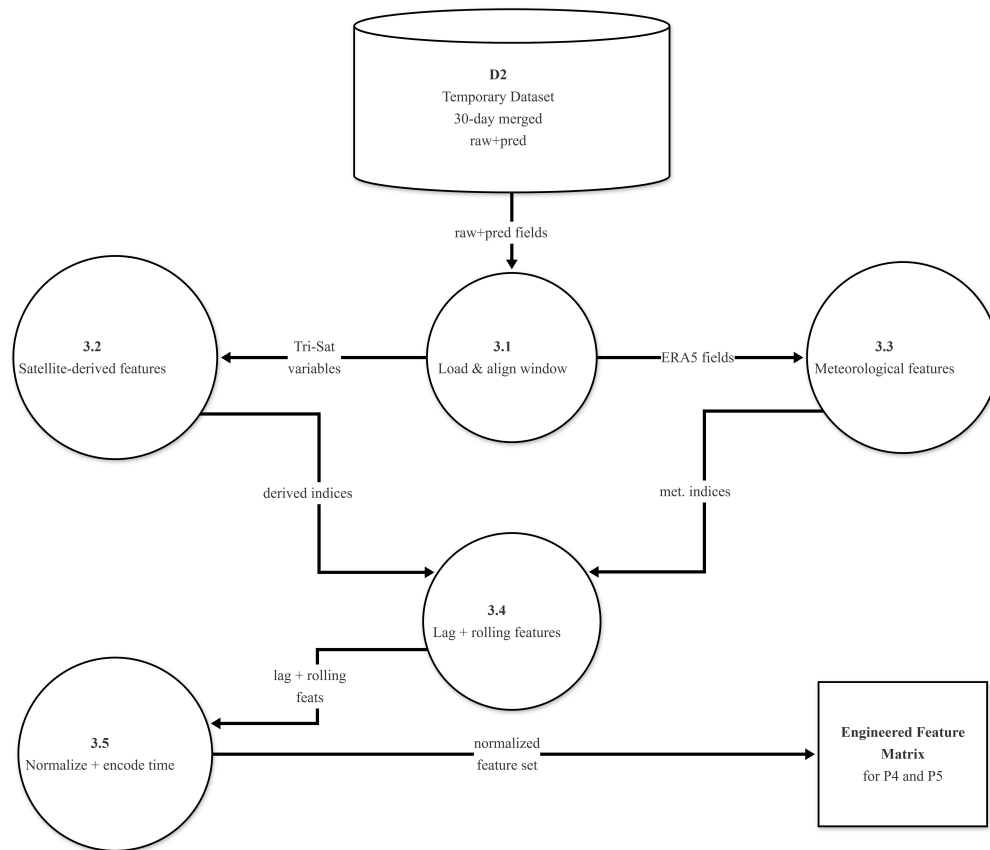


Figure 5.8: DFD - Level 2.3: Feature Engineering Module

Figure 5.8 shows how D2 is transformed into a structured feature matrix, including spectral indices, meteorological variables, temporal lags, rolling statistics, and cyclic calendar encodings.

DFD Level 2.4: Ensemble PM2.5 Prediction Model

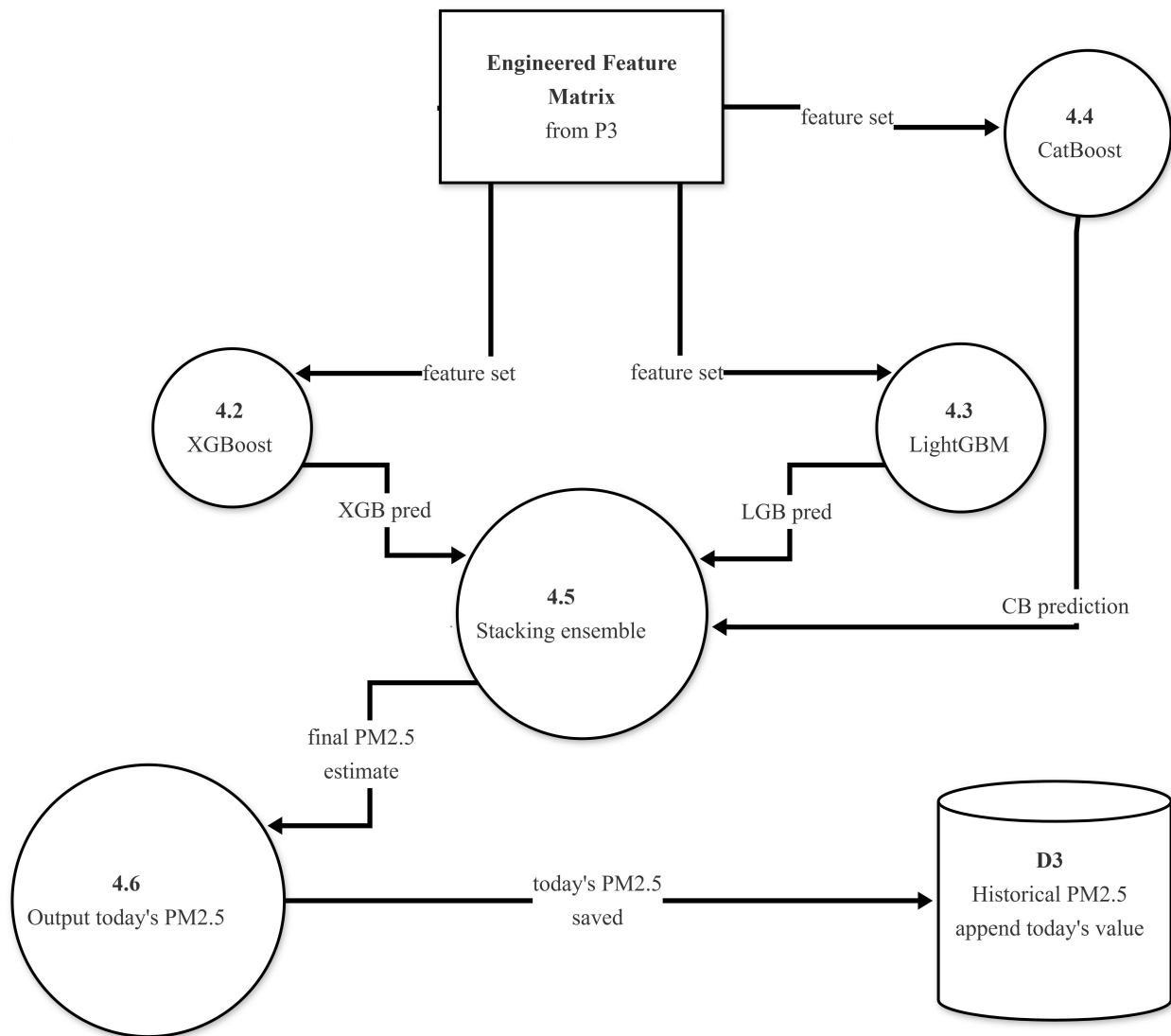


Figure 5.9: DFD - Level 2.4: Ensemble PM2.5 Prediction Model

Figure 5.9 illustrates the ensemble prediction process using XGBoost, LightGBM, and CatBoost, combined via stacking to produce daily PM_{2.5} estimates stored in D3.

DFD Level 2.5: LSTM PM2.5 Forecasting Model

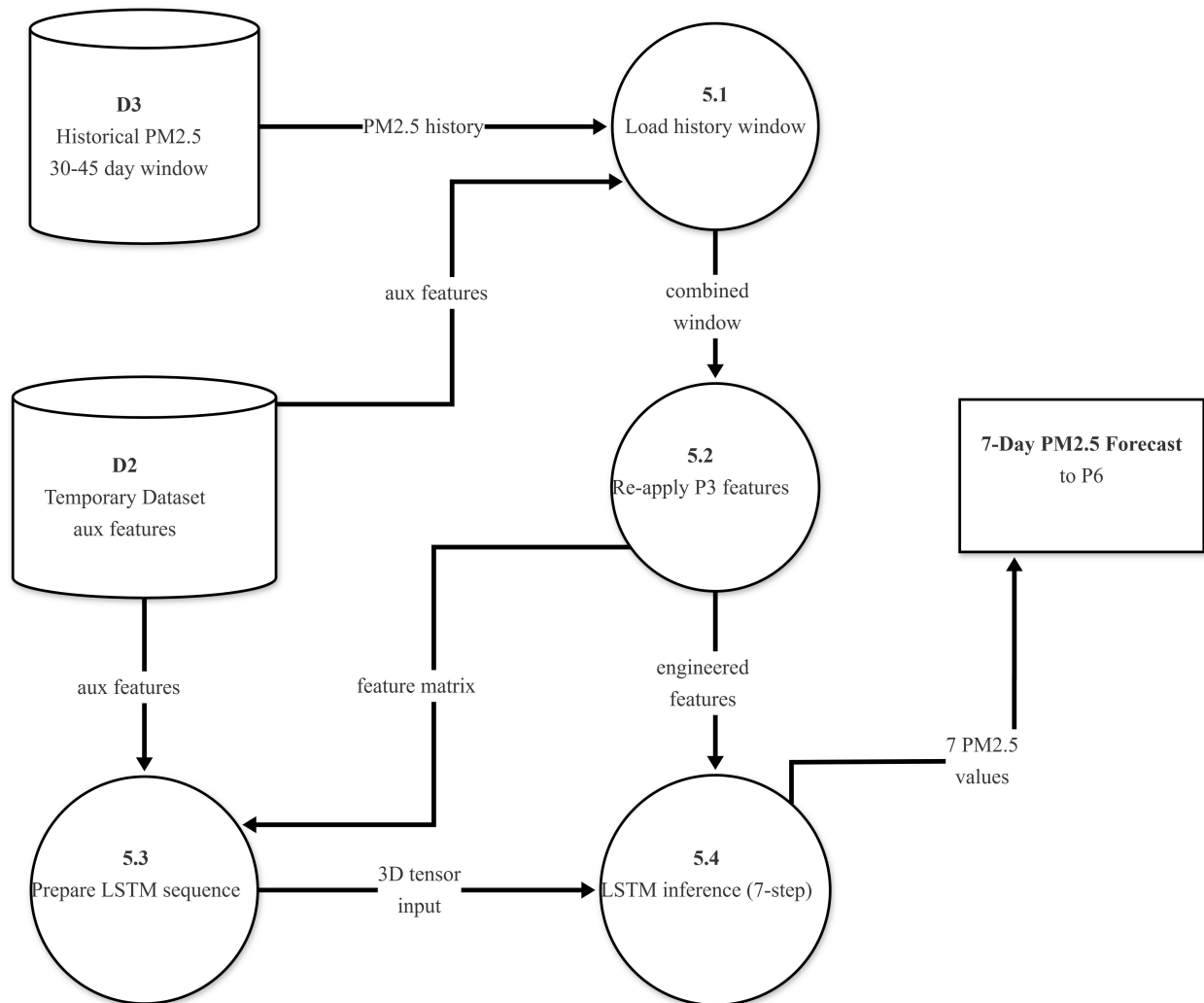


Figure 5.10: DFD - Level 2.5: LSTM PM2.5 Forecasting Model

Figure 5.10 shows the LSTM forecasting model, which produces a seven-day PM_{2.5} forecast using past predictions and corresponding features from D2.

DFD Level 2.6: Frontend Update Module

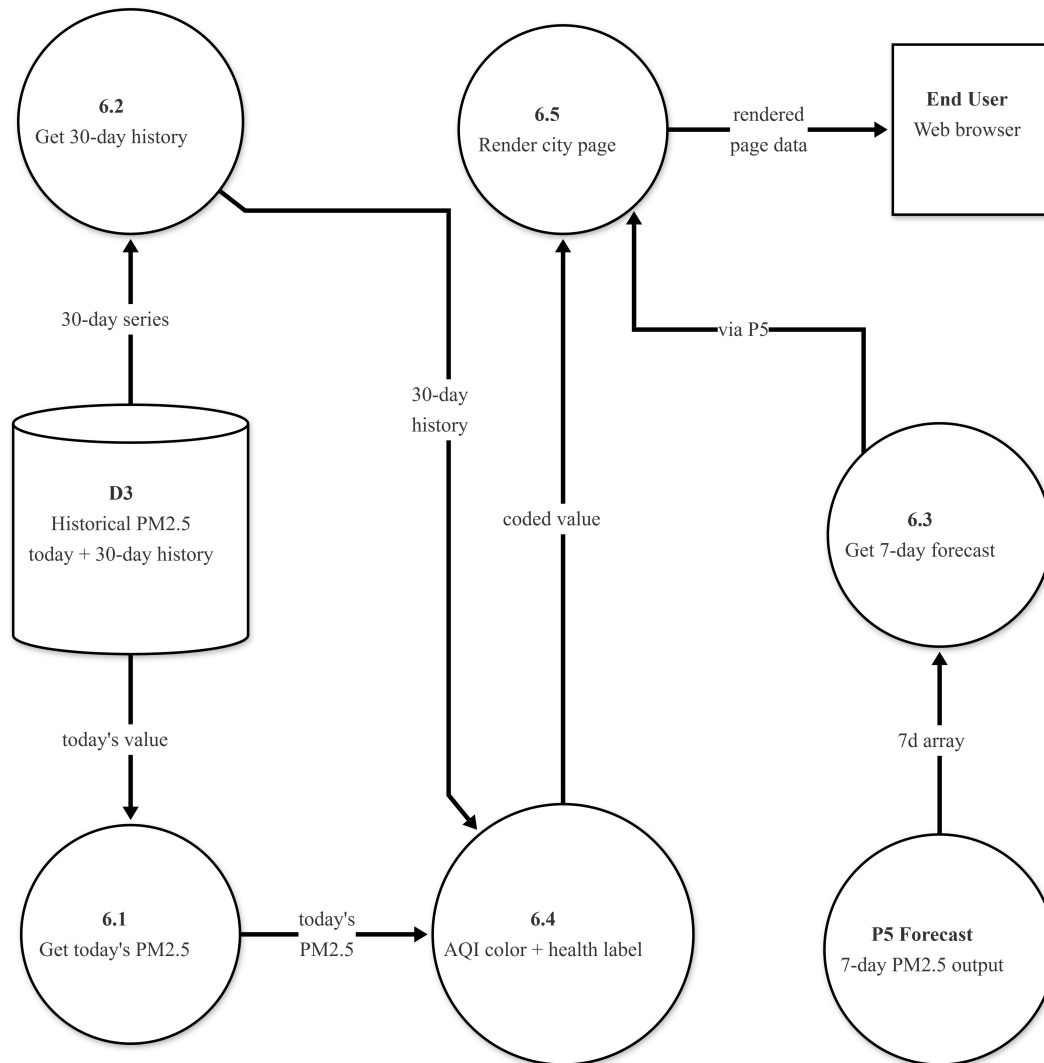


Figure 5.11: DFD - Level 2.6: Frontend Update Module

Figure 5.11 shows the Frontend Update Module (P6), which receives the PM_{2.5} daily predictions and seven-day forecasts and updates the web interface for each city.

6. Results

6.1 ERA5 Feature Forecasting

The LSTM forecaster was applied to generate a complete real-time set of ERA5 meteorological features. Table 6.1 summarises per-variable forecasting performance across the four deployment sites. Variables with high temporal persistence, such as pressure, temperature, and leaf area index, achieved R^2 above 0.98, while stochastic variables such as precipitation and surface runoff were comparatively less predictable.

Table 6.1: ERA5 variable forecasting performance using LSTM across deployment sites.

Variable	R^2	Accuracy (%)
Pressure	0.998	99.7
Leaf Area Index (High)	0.999	98.9
Temperature	0.988	97.2
Dew Point	0.986	90.8
Leaf Area Index (Low)	0.990	98.2
Thermal Radiation	0.984	98.5
Soil Water	0.977	95.5
Humidity	0.957	96.5
Evaporation	0.949	90.2
Boundary Layer Height	0.944	88.5
Wind Speed	0.934	98.5
Solar Radiation	0.933	92.2
Precipitation	0.931	72.5
Surface Water	0.910	77.5
Wind Direction (cos)	0.883	83.2
Wind Direction (sin)	0.869	81.9
Average	0.952	91.3

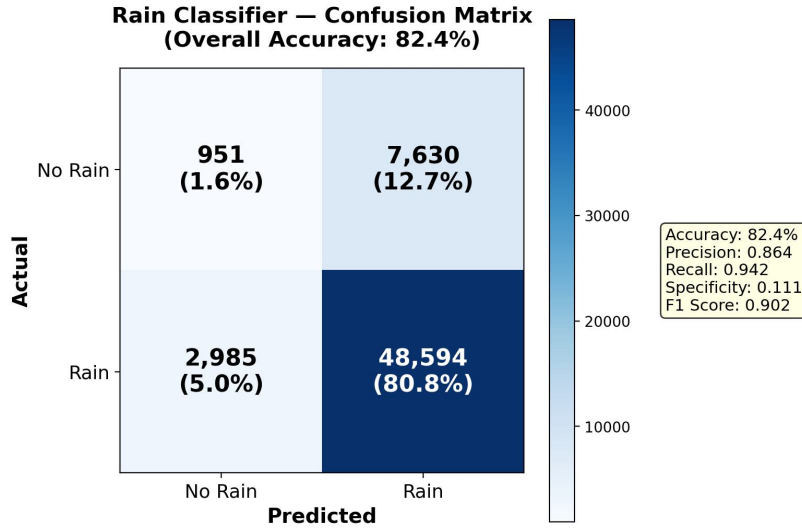


Figure 6.1: Confusion matrix for the XGBoost rain classifier.

Table 6.2: XGBoost rain classifier performance metrics.

Metric	Value
Accuracy	82.4%
Precision	0.864
Recall	0.942
Specificity	0.111
F1 Score	0.902

For precipitation specifically, a dedicated XGBoost rain classifier was used to support the LSTM prediction of rain amount, since rain occurrence is better treated as a binary classification problem. The confusion matrix for the classifier is shown in Figure 6.1, and the corresponding performance metrics are listed in Table 6.2.

6.2 MODIS AOD₅₅₀ Gap-Filling at Deployment Sites

MODIS AOD₅₅₀ observations at the four deployment sites contain frequent gaps due to cloud cover and retrieval failures. To ensure a continuous input for the downstream PM_{2.5} prediction model, a gap-filling model was trained to predict AOD₅₅₀ from ERA5 meteorological features. This model predicts AOD₅₅₀ only; it does not estimate PM_{2.5} directly.

Transfer learning models were applied to the deployment sites. Table 6.3 reports R² performance for both site-specific and transfer learning approaches.

Table 6.3: MODIS AOD₅₅₀ gap-filling performance at deployment sites.

Site	Site R ²	Transfer Learning R ²	Best Approach
Kathmandu	0.322	0.660	Transfer Learning
Pokhara	0.335	0.547	Transfer Learning
Chitwan	0.504	0.506	Transfer Learning
Birgunj	0.685	0.440	Site-Specific

Transfer learning improved gap-filling performance at three of the four sites. Site-level AOD₅₅₀ variability and observation gap rates are summarised in Table 6.4.

Table 6.4: AOD₅₅₀ distribution and data gap statistics at deployment sites.

Site	Mean AOD ₅₅₀	Std AOD ₅₅₀	Gap (%)
Kathmandu	0.322	0.207	38
Pokhara	0.335	0.263	36
Chitwan	0.504	0.344	36
Birgunj	0.685	0.385	44

6.3 Sentinel-2 Band Prediction

Sentinel-2 reflectance bands were predicted from ERA5 features to provide gap-filled surface observations. Per-band prediction performance (R²) is shown in Figure 6.2.

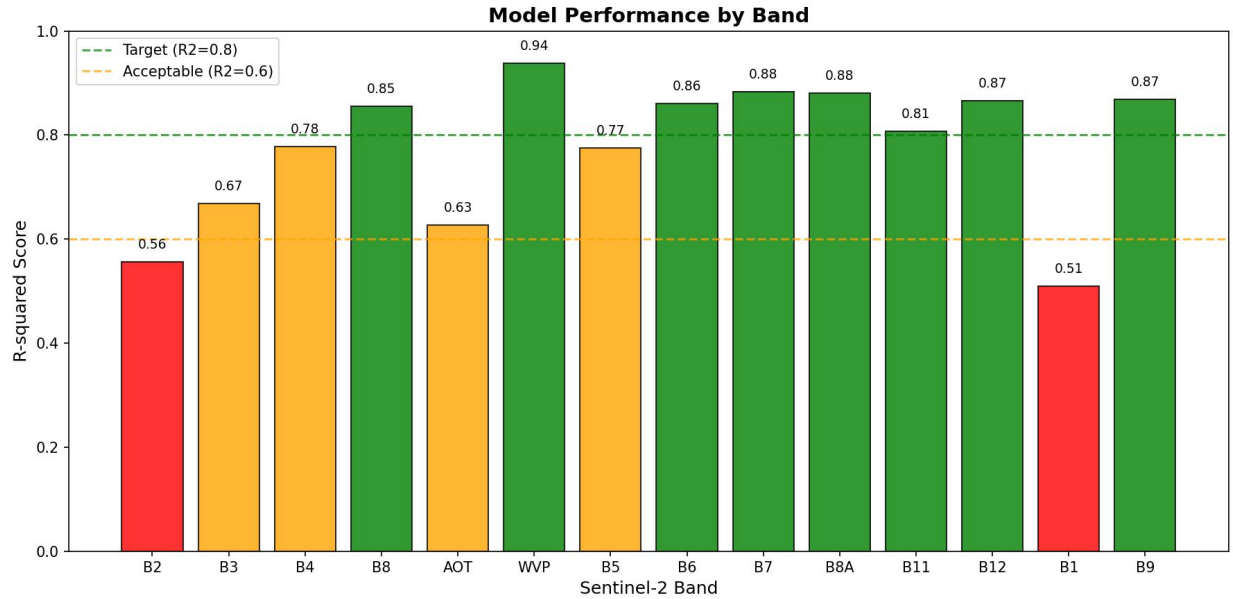


Figure 6.2: Sentinel-2 per-band prediction performance (R²) using ERA5-based gap-filling.

Bands in the NIR and SWIR ranges were consistently more predictable compared to visible and aerosol-sensitive bands.

6.4 Sentinel-5P Trace Gas Prediction

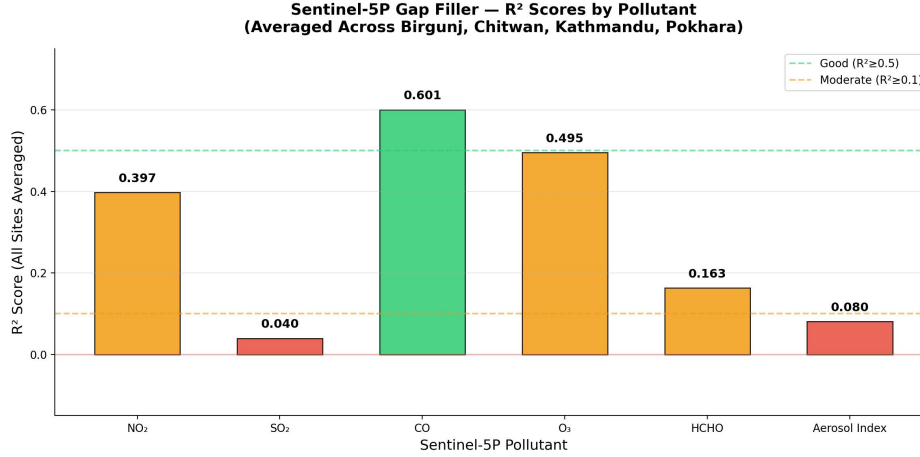


Figure 6.3: Sentinel-5P feature prediction performance (average R² across features).

Sentinel-5P trace gas columns (CO, NO₂, O₃, HCHO, AI, SO₂) were also predicted from ERA5 features. Average R² across all gases is shown in Figure 6.3.

Table 6.5: Sentinel-5P poorly predicted trace gas summary statistics.

Gas	Mean	Data Gap (%)
HCHO	0.00019	29
AI	−0.728	0.45
SO ₂	0.00031	63

CO and O₃ showed moderate predictability, while HCHO, AI, and SO₂ performed poorly. Summary statistics for the three lowest-performing gases are provided in Table 6.5.

6.5 Ensemble PM_{2.5} Prediction

The base ensemble model was trained using ERA5, gap-filled MODIS AOD₅₅₀, Sentinel-2, and Sentinel-5P features, and evaluated on a chronological held-out test set. The scatter plot for the base model is shown in Figure 6.4, and performance metrics are listed in Table 6.6.

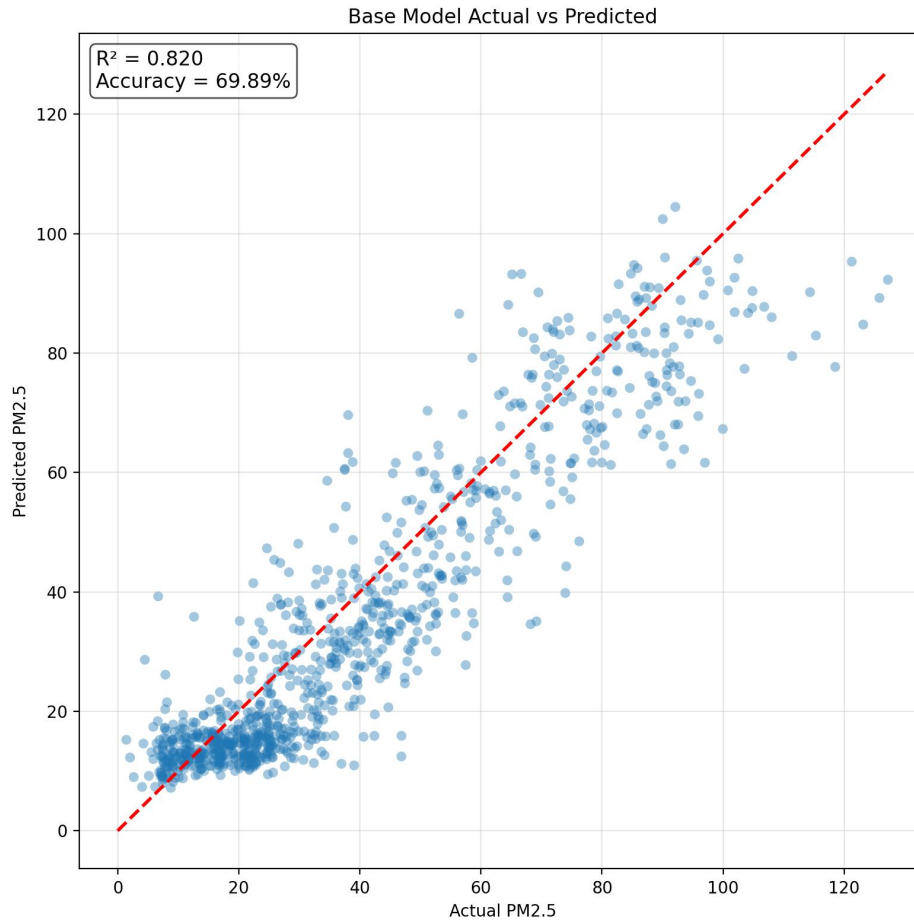


Figure 6.4: Ensemble base model: actual vs predicted $\text{PM}_{2.5}$.

Table 6.6: Ensemble $\text{PM}_{2.5}$ model performance before and after deployment-site fine-tuning.

Metric	Base Model	Fine-Tuned
R^2	0.8201	0.8909
RMSE ($\mu\text{g}/\text{m}^3$)	11.09	—
MAE ($\mu\text{g}/\text{m}^3$)	8.64	—
Accuracy (%)	69.89	79.83

When adapted using deployment-site data, performance improved to $R^2 = 0.8909$ and accuracy = 79.83%, evaluated on a held-out deployment split. The gain suggests the model benefits substantially from local fine-tuning, as site-specific patterns in $\text{PM}_{2.5}$ sources and seasonal behaviour are better captured once local sensor data is incorporated.

Per-site predictions after fine-tuning are shown in Figure 6.5.

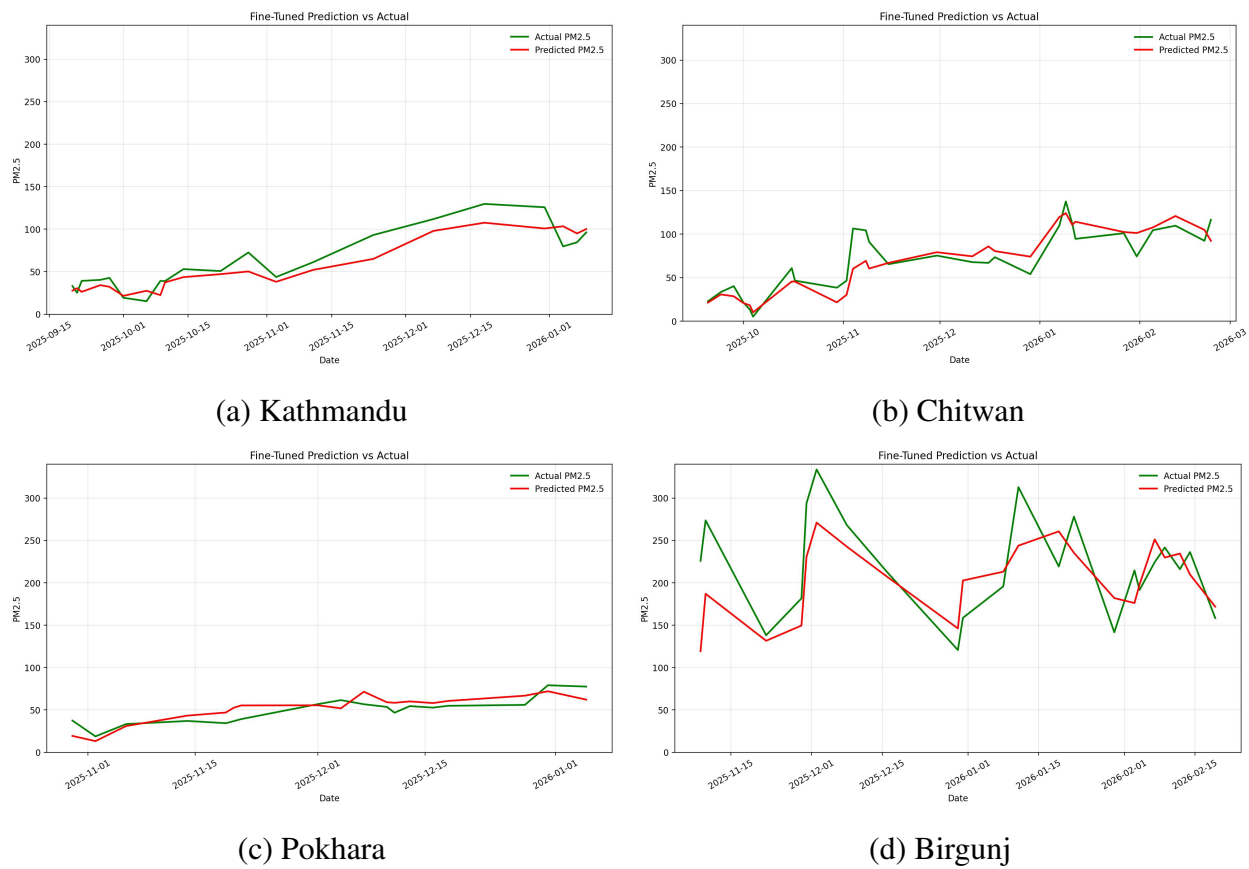


Figure 6.5: Per-site ensemble $PM_{2.5}$ predictions after fine-tuning

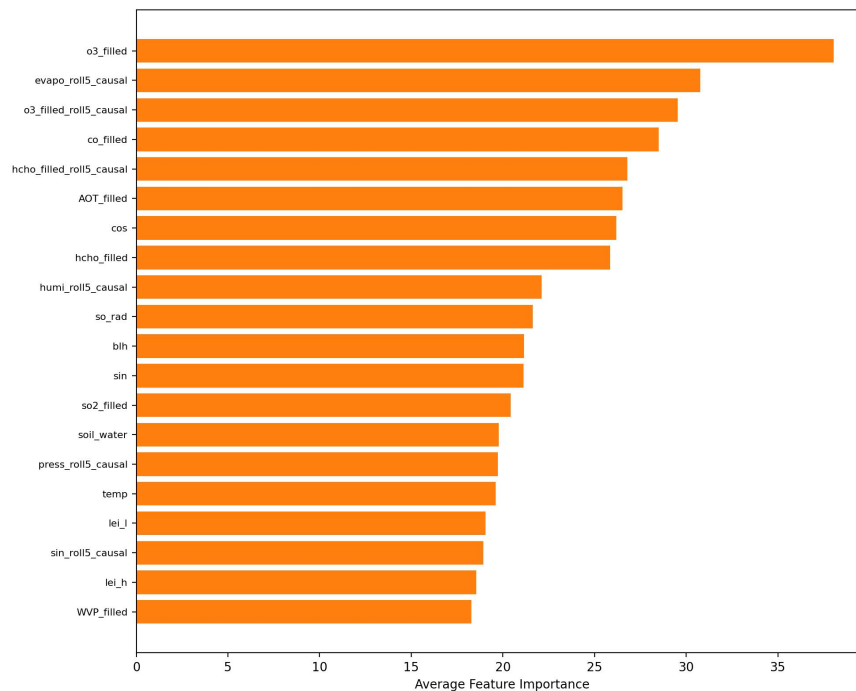


Figure 6.6: Top 20 feature importance values for the Ensemble $PM_{2.5}$ Prediction Model.

Figure 6.6 shows the top 20 features driving the Ensemble PM_{2.5} Prediction Model, highlighting the impact of both satellite measurements as well as meteorological analysis.

6.6 LSTM PM_{2.5} Forecasting

A separate LSTM model was trained for 7-day ahead PM_{2.5} forecasting using the gap-filled ERA5 and satellite features. Overall performance metrics are listed in Table 6.7.

Table 6.7: LSTM 7-day PM_{2.5} forecasting performance.

Metric	Value
R ²	0.8335
RMSE ($\mu\text{g}/\text{m}^3$)	19.18
MAE ($\mu\text{g}/\text{m}^3$)	9.87
Accuracy (%)	80.75

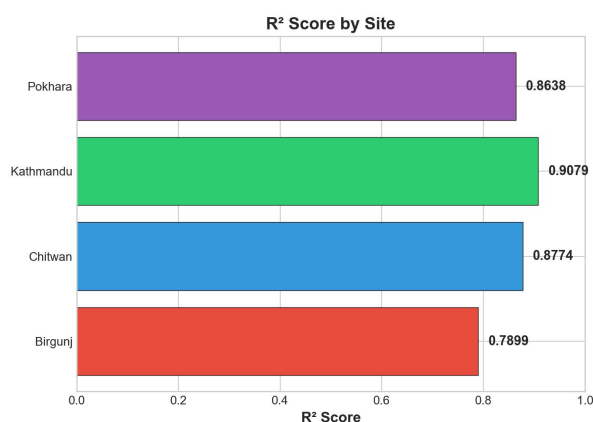


Figure 6.7: Site-wise R² of the LSTM PM_{2.5} forecaster.

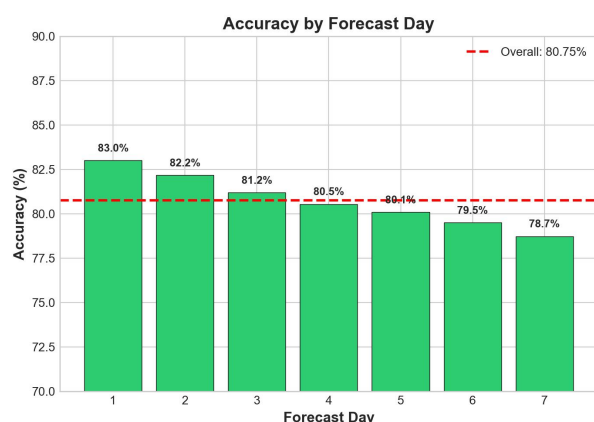


Figure 6.8: LSTM forecast accuracy across the 7-day prediction horizon.

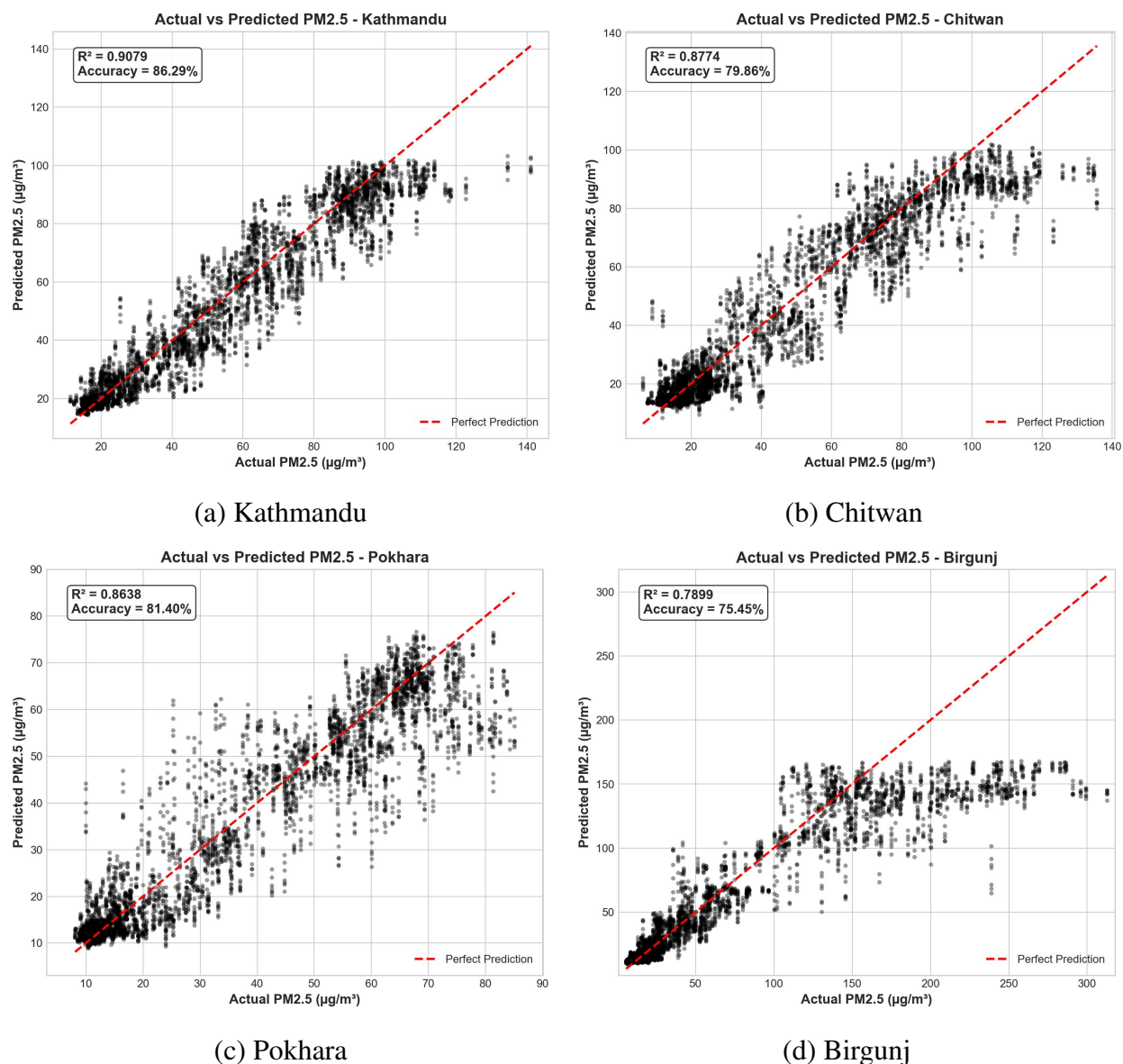


Figure 6.9: Per-site LSTM PM_{2.5} actual vs predicted scatter plots

Figure 6.9 shows the relationship between observed and predicted PM_{2.5} values at each site. Kathmandu and Chitwan show tighter clustering around the perfect prediction line, consistent with their more stable seasonal pollution patterns. Birgunj shows the widest scatter, particularly at high PM_{2.5} values, which is likely attributable to episodic cross-border pollution transport events that are difficult to capture from meteorological features alone.

7. Discussion

7.1 ERA5 Feature Forecasting

ERA5-based feature forecasting performed well overall, with an average R^2 of 0.952 across all variables (Table 6.1). Variables such as pressure, temperature, and leaf area index are driven by strong seasonal and diurnal cycles, which the LSTM captures effectively over long sequences. The lower performance for precipitation, surface water, and wind direction is expected since these are inherently stochastic and spatially heterogeneous.

The XGBoost rain classifier achieved a reasonable overall accuracy of 82.4% and a high recall of 0.942, but specificity was very low at 0.111. This is a direct consequence of the class imbalance in the training data, where rain events (51,579 samples) vastly outnumber no-rain events (8,581 samples), as summarised in Table 7.1. The large spatial footprint of ERA5 grid cells (roughly 11 km) also means that rain is flagged across a wide area even for localised events, further biasing the classifier toward predicting rain. The high recall does ensure that actual rain events are rarely missed, which is the more practically relevant outcome for air quality forecasting.

Table 7.1: Class distribution in the XGBoost rain classifier training data.

Class	Samples	Share (%)
Rain	51,579	85.7
No Rain	8,581	14.3
Total	60,160	100.0

7.2 MODIS AOD₅₅₀ Gap-Filling

MODIS AOD₅₅₀ retrievals at all four sites are affected by substantial data gaps, ranging from 36% at Pokhara and Chitwan up to 44% at Birgunj (Table 6.4). These gaps arise from cloud cover and aerosol retrieval failures, and they make it impractical to use raw MODIS observations directly for near-real-time PM_{2.5} prediction. The gap-filling model addresses this by predicting AOD₅₅₀ from ERA5 meteorological features whenever a satellite observation is unavailable.

Transfer learning improved performance at three of the four sites, which shows the benefit of sharing information across locations when site-specific training data is limited. Birgunj is the exception: its site-specific R^2 of 0.685 is notably higher than at the other three sites, meaning a locally trained model already has enough signal to outperform the transfer approach.

The remaining variability at sites like Pokhara and Kathmandu (standard deviations of 0.263 and 0.207 respectively) reflects localised aerosol dynamics that are difficult to capture from ERA5 alone. Pokhara in particular sits in a valley, where topographic channelling of winds and local moisture patterns create conditions that are not well represented by the coarser ERA5 grid.

7.3 Sentinel-2 Band Prediction

NIR and Red Edge bands (B5 to B8A) and SWIR bands (B9, B11, B12) showed the strongest predictability, with average R^2 values of 0.859 and 0.847 respectively (Figure 6.2). These bands are closely tied to vegetation and soil moisture signals, both of which are reasonably well captured by ERA5 variables such as soil water content, leaf area index, and evaporation.

Visible bands (B1 to B4) and the aerosol optical thickness band (AOT) were considerably harder to predict, with average R^2 around 0.627 for both groups. Visible reflectance is strongly influenced by atmospheric scattering, and ERA5 does not provide a direct aerosol loading variable. Without that, the model cannot fully reconstruct the aerosol-driven variability in these bands. The water vapour band (WVP) performed best of all at $R^2 = 0.938$, which is not surprising given its direct correspondence to ERA5 humidity and dew point.

7.4 Sentinel-5P Trace Gas Prediction

CO and O₃ showed moderate predictability because their atmospheric concentrations are meaningfully coupled to meteorological conditions such as temperature, boundary layer height, and wind, all of which are available in ERA5. The remaining gases performed poorly (Table 6.5). HCHO and SO₂ both returned near-zero R^2 values, while the aerosol index (AI) produced a strongly negative mean value (−0.728). These gases are characterised by short atmospheric lifetimes, episodic or point-source emission patterns, and high spatial variability, none of which can be reconstructed from ERA5 meteorology alone. The large data gap for SO₂ (63%) also means the available training samples are sparse and potentially unrepresentative.

Given these results, the contribution of Sentinel-5P features to the ensemble PM_{2.5} model is primarily through the more predictable gases, while the poorly predicted gases add limited useful signal.

7.5 Ensemble PM_{2.5} Prediction

The base ensemble model achieved an R^2 of 0.8201, demonstrating that a reasonable PM_{2.5} prediction is possible even before any site-specific calibration. The improvement after fine-tuning, with R^2 rising to 0.8909 and accuracy from 69.89% to 79.83% (Table 6.6), confirms that local sensor data carries information that the generalised base model cannot fully capture. This is consistent with the known site-to-site variability in PM_{2.5} sources across Nepal: Kathmandu is dominated by traffic and biomass burning, Birgunj by industrial and cross-border transport, and Chitwan and

Pokhara by a mix of agricultural and domestic sources.

The per-site scatter plots in Figure 6.5 show that prediction quality varies across sites, with the model performing better at sites that have more stable seasonal patterns in their $\text{PM}_{2.5}$ time series.

7.6 LSTM $\text{PM}_{2.5}$ Forecasting

The LSTM forecaster achieved an overall R^2 of 0.8335 and an accuracy of 80.75% over a 7-day prediction horizon (Table 6.7), which is a competitive result for multi-day air quality forecasting at these scales. The RMSE of $19.18 \mu\text{g}/\text{m}^3$ is higher than the ensemble model's $11.09 \mu\text{g}/\text{m}^3$, which is expected since forecasting further ahead introduces compounding uncertainty from the ERA5 feature forecasts.

The forecast accuracy as a function of lead day, shown in Figure 6.8, illustrates how performance degrades with increasing horizon, which is a natural consequence of error accumulation in the input feature forecasts. Despite this, the model maintains reasonable accuracy even at 7 days, suggesting that the dominant $\text{PM}_{2.5}$ patterns in these locations are driven by persistent meteorological conditions rather than sudden episodic events.

7.7 Overall Assessment

Taken together, the results demonstrate that the integrated framework is capable of producing operationally useful $\text{PM}_{2.5}$ predictions and 7-day forecasts across four diverse monitoring sites in Nepal. The combination of ERA5-based feature forecasting, transfer-learning-based gap-filling for MODIS AOD_{550} , and selective use of Sentinel-2 and Sentinel-5P features provides a robust input pipeline that handles the missing data problem inherent in satellite-based monitoring. Fine-tuning the ensemble model with local sensor data is a key step that translates general predictive skill into site-level accuracy, and is recommended as a standard step whenever deploying this framework at a new location.

8. Conclusions

A near real-time $\text{PM}_{2.5}$ prediction and forecasting framework for Nepal has been developed, integrating MODIS, Sentinel-2, and Sentinel-5P satellite observations with ERA5 reanalysis data and ground-based measurements through ensemble machine learning and LSTM-based deep learning models.

The results demonstrate that satellite-derived features, when combined with meteorological reanalysis data and systematic gap-filling, can reliably estimate daily $\text{PM}_{2.5}$ concentrations without depending on dense ground sensor networks. The base ensemble model achieved $R^2 = 0.8201$ on a chronological test set, improving to $R^2 = 0.8909$ after deployment-site fine-tuning, confirming that local calibration is critical for site-level accuracy. The LSTM forecaster achieved $R^2 = 0.8335$ and 80.75% accuracy over a 7-day horizon, with accuracy declining gradually with lead time as expected from error accumulation in the input feature forecasts.

Transfer learning for MODIS AOD_{550} gap-filling improved performance at three of the four deployment sites, demonstrating that shared atmospheric patterns across locations can be leveraged effectively when local training data is limited. The modular pipeline design ensures reproducibility and operational continuity, with the system running automatically each day for four cities: Kathmandu, Pokhara, Chitwan, and Birgunj.

Overall, the framework demonstrates that satellite-based $\text{PM}_{2.5}$ monitoring is a viable and scalable alternative to ground infrastructure in data-sparse regions like Nepal, and can support public health and environmental policy decisions where conventional monitoring is unavailable or unreliable.

9. Limitations & Future Enhancements

9.1 Limitations

- Ground sensor availability was limited to five sites in Nepal and three in India, which restricts the spatial diversity of training data and may introduce systematic bias at locations with pollution characteristics not well represented during training.
- Satellite observations contain frequent daily gaps due to cloud cover and retrieval failures, necessitating gap-filling models that introduce additional uncertainty compared to direct measurements.
- Deployment testing was restricted to four recently instrumented sites, limiting the spatial generalizability of the results to other regions or terrain types within Nepal.
- While the models capture temporal pollution patterns reasonably well, absolute $\text{PM}_{2.5}$ magnitudes at new deployment locations may be systematically underestimated, as local emission characteristics and aerosol properties that differ from training sites cannot be fully inferred from satellite and reanalysis inputs alone. Co-located ground measurements are recommended to correct this bias before operational deployment.

9.2 Future Enhancements

- Expansion to additional deployment sites across Nepal and neighbouring regions to improve spatial coverage and model generalizability.
- Incorporation of hourly or sub-daily satellite and sensor data for higher temporal resolution $\text{PM}_{2.5}$ estimates and forecasts.
- Integration of low-cost sensors and crowd-sourced air quality data to support local bias correction at new deployment sites with minimal infrastructure investment.
- Exploration of spatio-temporal deep learning architectures to jointly model pollution dynamics across multiple sites, potentially reducing the dependence on site-specific calibration.
- Potential adaptation of the framework for other South Asian countries or regions facing similar data sparsity and monitoring infrastructure challenges.

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Appendices

Appendix A — Model Hyperparameters

a. Ensemble PM_{2.5} Prediction Model

Base Learner Hyperparameters

1. XGBoost (XGBRegressor)

- `n_estimators = 758`
- `learning_rate = 0.043688`
- `max_depth = 3`
- `min_child_weight = 14`
- `subsample = 0.899283`
- `colsample_bytree = 0.794906`
- `reg_alpha = 5.668891`
- `reg_lambda = 0.003127`
- `tree_method = hist`
- `random_state = 42`

2. LightGBM (LGBMRegressor)

- `n_estimators = 390`
- `learning_rate = 0.132721`
- `max_depth = 4`
- `num_leaves = 54`
- `min_child_samples = 64`
- `subsample = 0.511630`
- `colsample_bytree = 0.546173`
- `reg_alpha = 3.995919`
- `reg_lambda = 0.007631`
- `random_state = 42`
- `verbose = -1`

3. CatBoost (CatBoostRegressor)

- `iterations = 690`
- `learning_rate = 0.100041`
- `depth = 7`
- `l2_leaf_reg = 2.887049`
- `bagging_temperature = 7.403328`
- `random_strength = 0.242500`
- `border_count = 56`
- `random_seed = 42`
- `verbose = False`

b. Stacking / Ensemble Setup

- Final estimator: `Ridge(alpha=1.0, solver='auto', fit_intercept=True)`
- Base model predictions are inputs to the Ridge meta-model.
- `passthrough = False`, so original features are not passed to the final model.

LSTM PM_{2.5} Forecasting Model

Hyperparameters

4. LSTM

- `hidden_size = 256`
- `num_layers = 2`
- `dropout = 0.173`
- `learning_rate = 0.00113`
- `batch_size = 32`
- `weight_decay = 0.0002`

Appendix B — Feature List

Ensemble PM_{2.5} Model Features

1. aod550_filled	17. o3_filled	33. t_rad
2. AOT_filled	18. so2_filled	34. temp
3. B1_filled	19. WVP_filled	35. wspeed
4. B11_filled	20. blh	36. hcho_filled_roll5_causal
5. B12_filled	21. cos	37. o3_filled_roll5_causal
6. B2_filled	22. dew	38. WVP_filled_roll5_causal
7. B3_filled	23. evapo	39. evapo_roll5_causal
8. B4_filled	24. humi	40. humi_roll5_causal
9. B5_filled	25. lei_h	41. press_roll5_causal
10. B6_filled	26. lei_l	42. sin_roll5_causal
11. B7_filled	27. preci	43. aod550_filled_mul_cos
12. B8_filled	28. press	44. aod550_filled_div_cos
13. B8A_filled	29. sin	45. cos_div_aod550_filled
14. co_filled	30. so_rad	46. aod550_filled_mul_evapo
15. hcho_filled	31. soil_water	47. aod550_filled_div_surf_water
16. no2_filled	32. surf_water	

LSTM Model Features

- All 35 basic features (Non-Engineered) from Ensemble Model's feature list.
- Past PM_{2.5} values (last 30 days) as input sequences

Appendix C — Dataset Details

- **Data Period:** 2017–2025
- **Temporal Resolution:** Daily
- **Spatial Resolution:**
 - Satellites: 5000 m and 600 m
 - ERA5: 11132 m
 - Ground sensors: As provided by OpenAQ
- **Ground Truth Source:** OpenAQ
- **Satellite / ERA5 Platform:** Google Earth Engine
- **Number of Training Sites:** 8
- **Number of Deployment Sites:** 4